

Retrocausal capacity of a quantum channel

Kaiyuan Ji (Cornell), **Seth Lloyd** (MIT), Mark M. Wilde (Cornell)



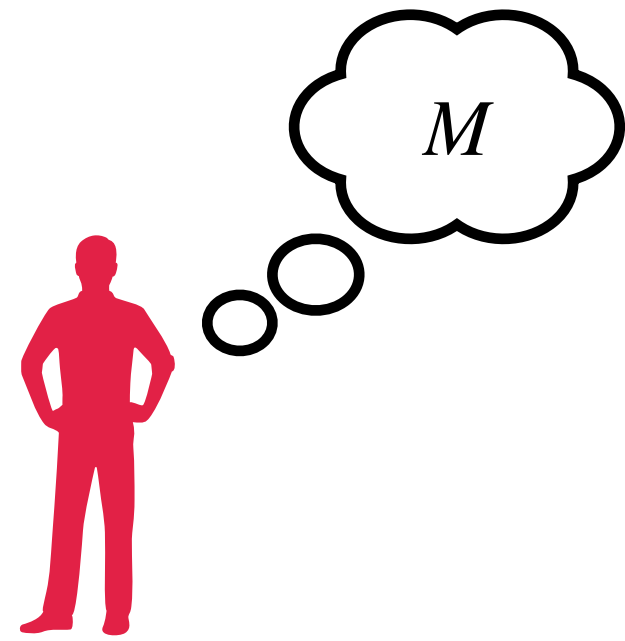
[arXiv:2509.08965](https://arxiv.org/abs/2509.08965)



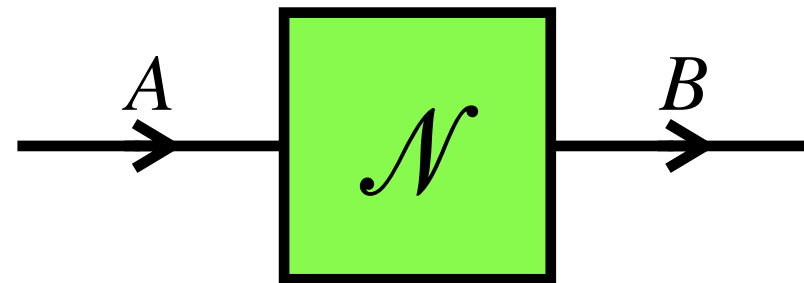
Quantum Resources 2026, Tokyo

Communication

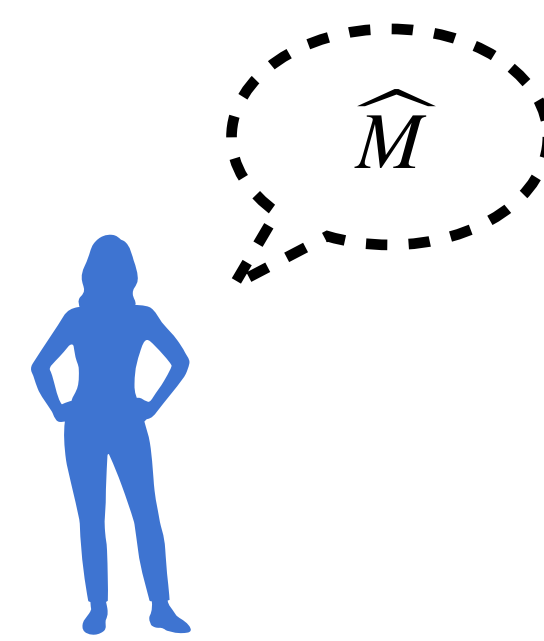
sender (past)



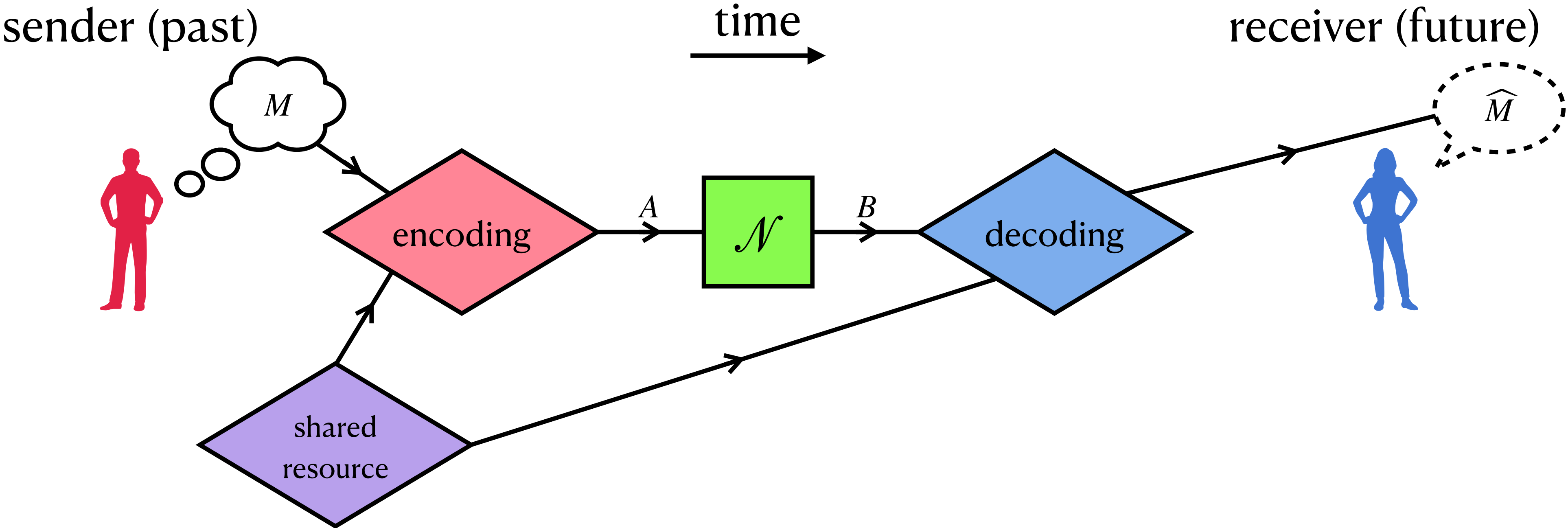
time



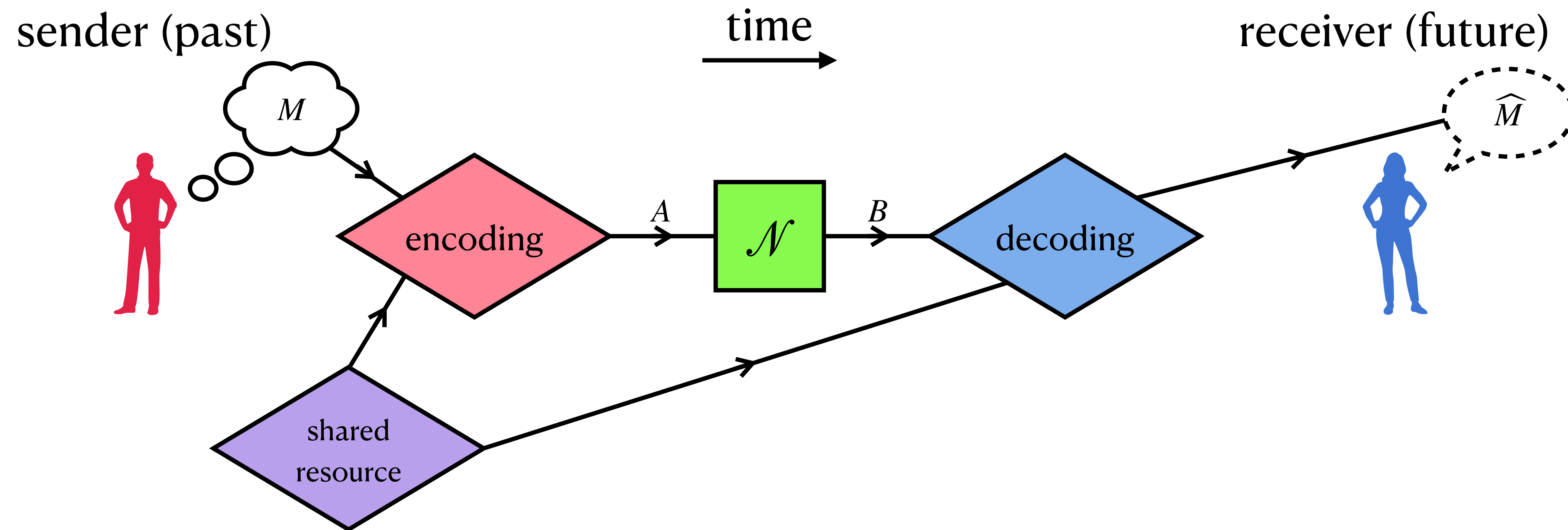
receiver (future)



Communication

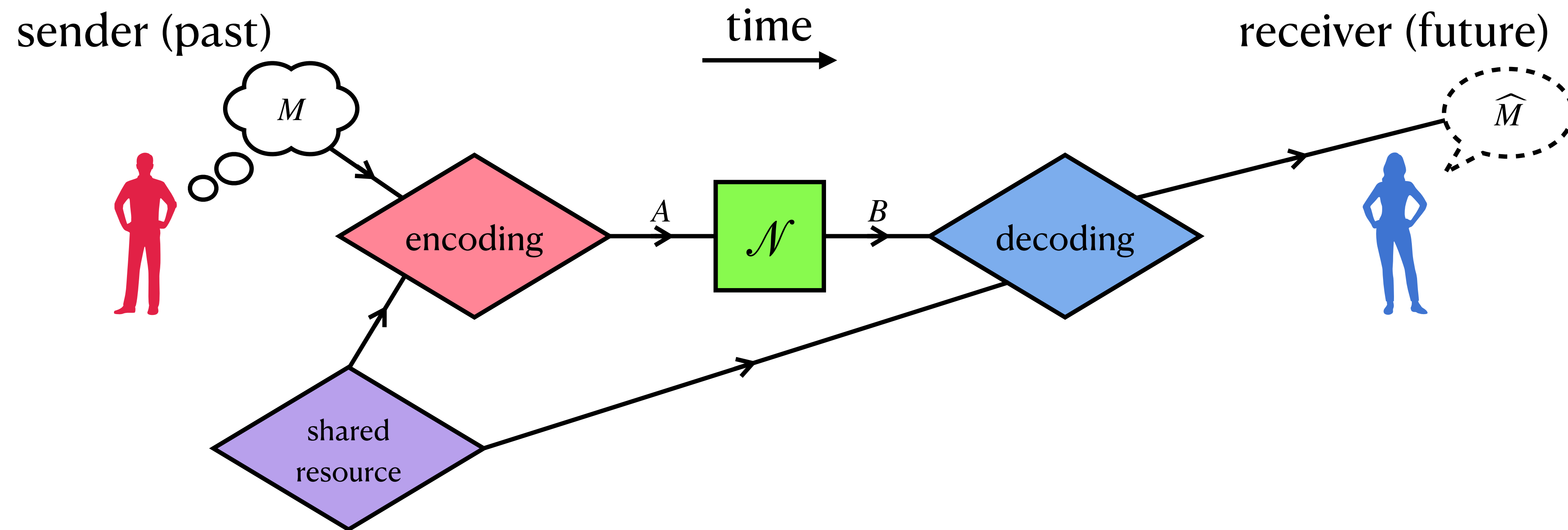


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Quantum Shannon theory studies the fundamental limit of communication (forward in time) through a given noisy channel (CPTP map) $\mathcal{N}_{A \rightarrow B}$

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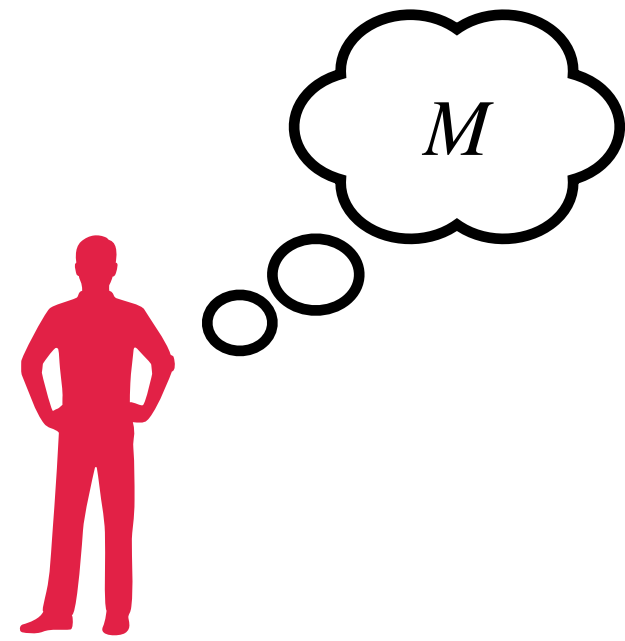
- “Capacity” of $\mathcal{N}_{A \rightarrow B}$

Holevo, Probl. Peredachi Inf. 9, 3 (1973)
Schumacher, PRA 54, 2614 (1996)

...
...

Communication

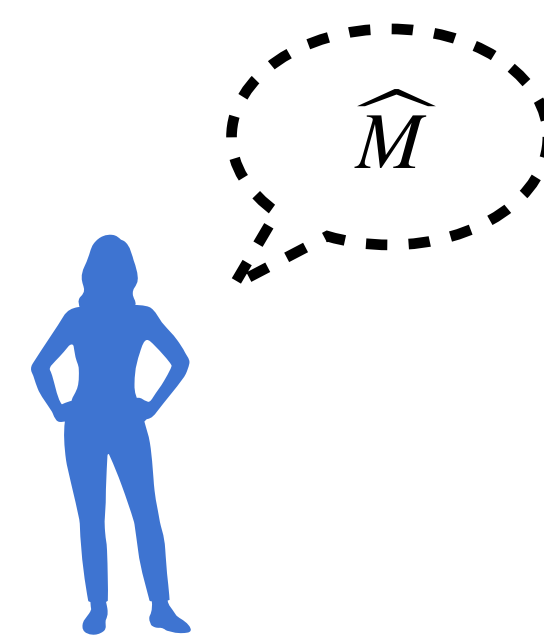
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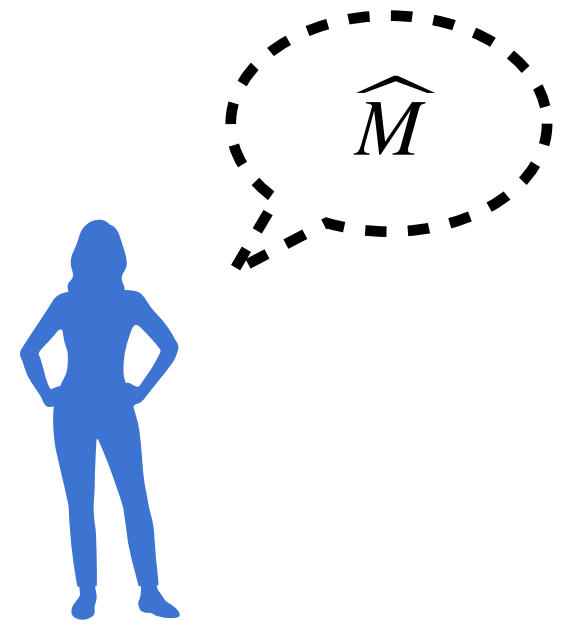


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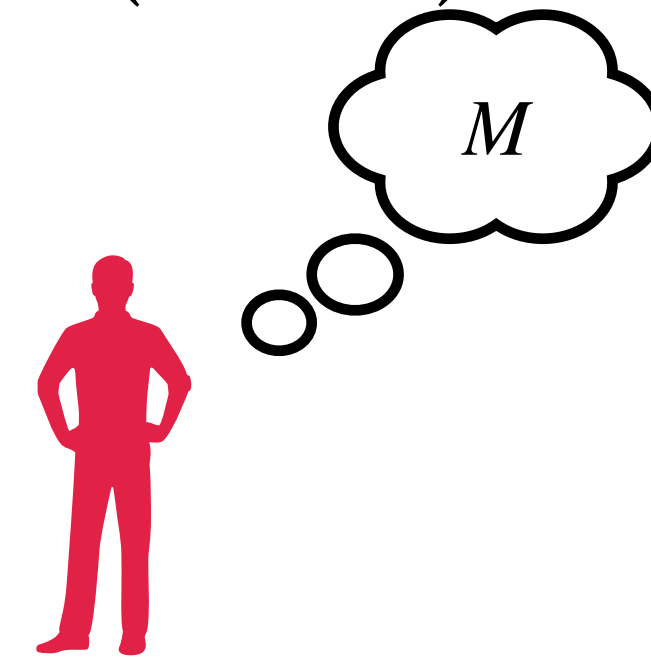
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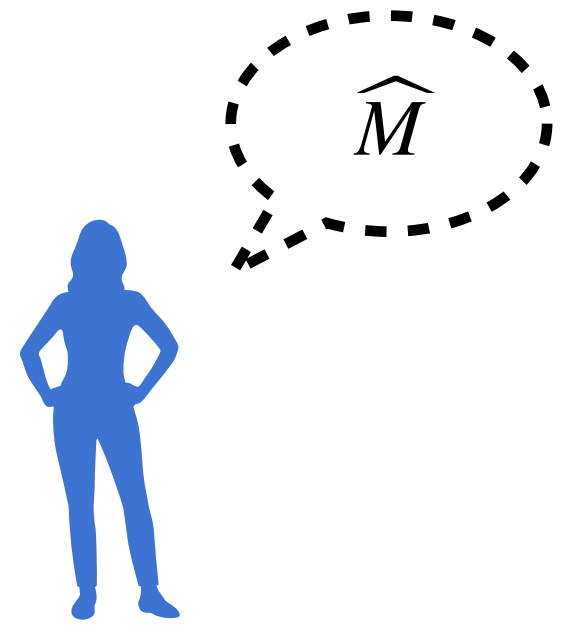


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Communication, but backward in time!

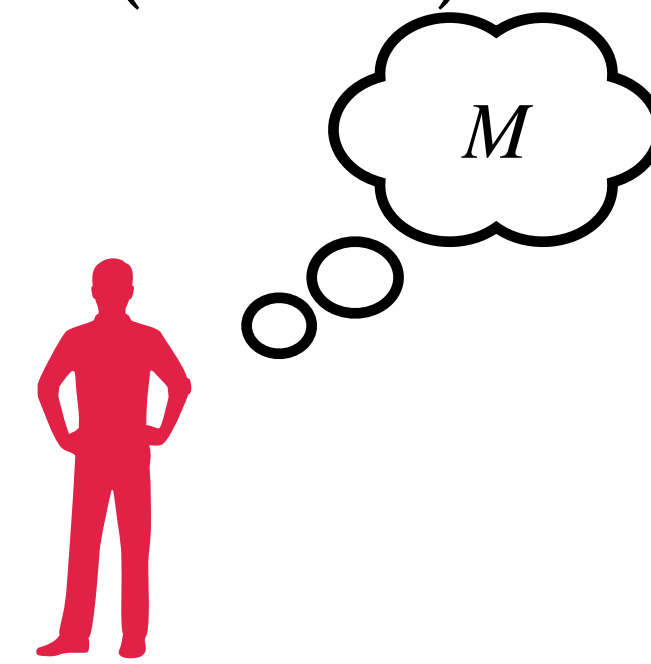
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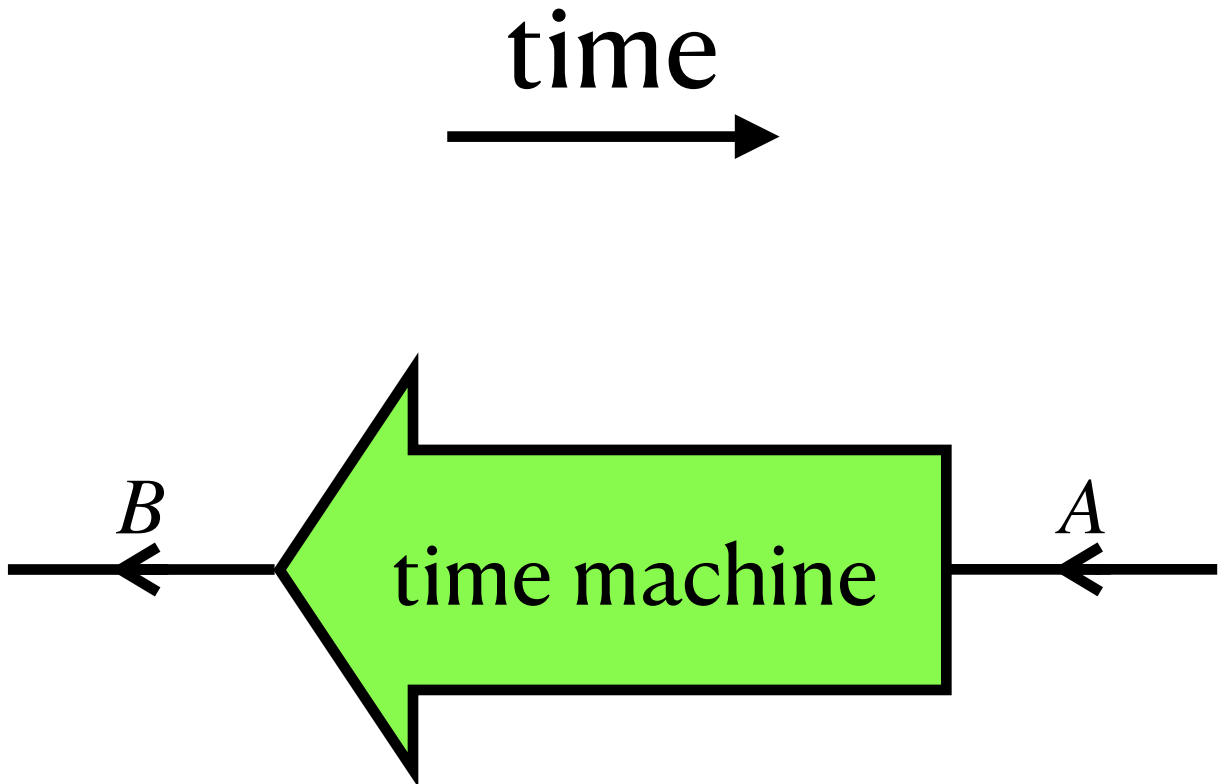
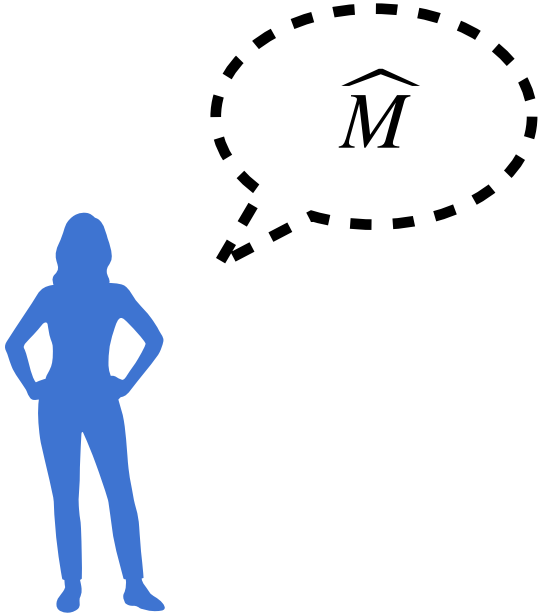


sender (future)

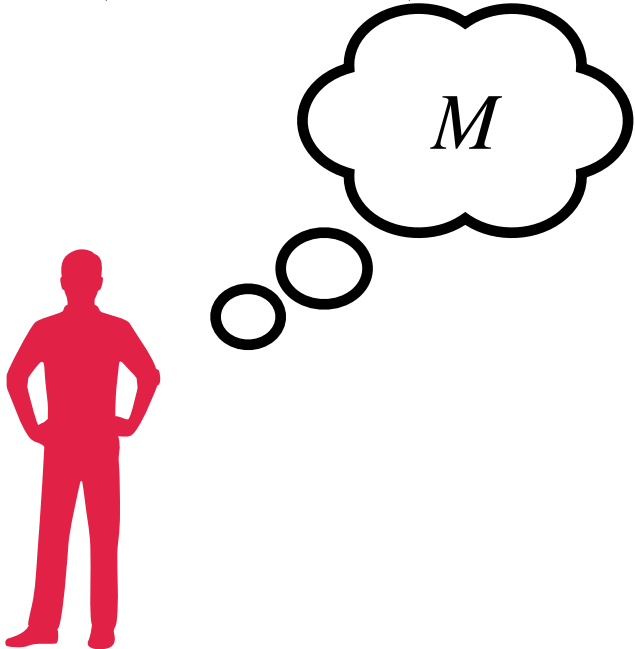


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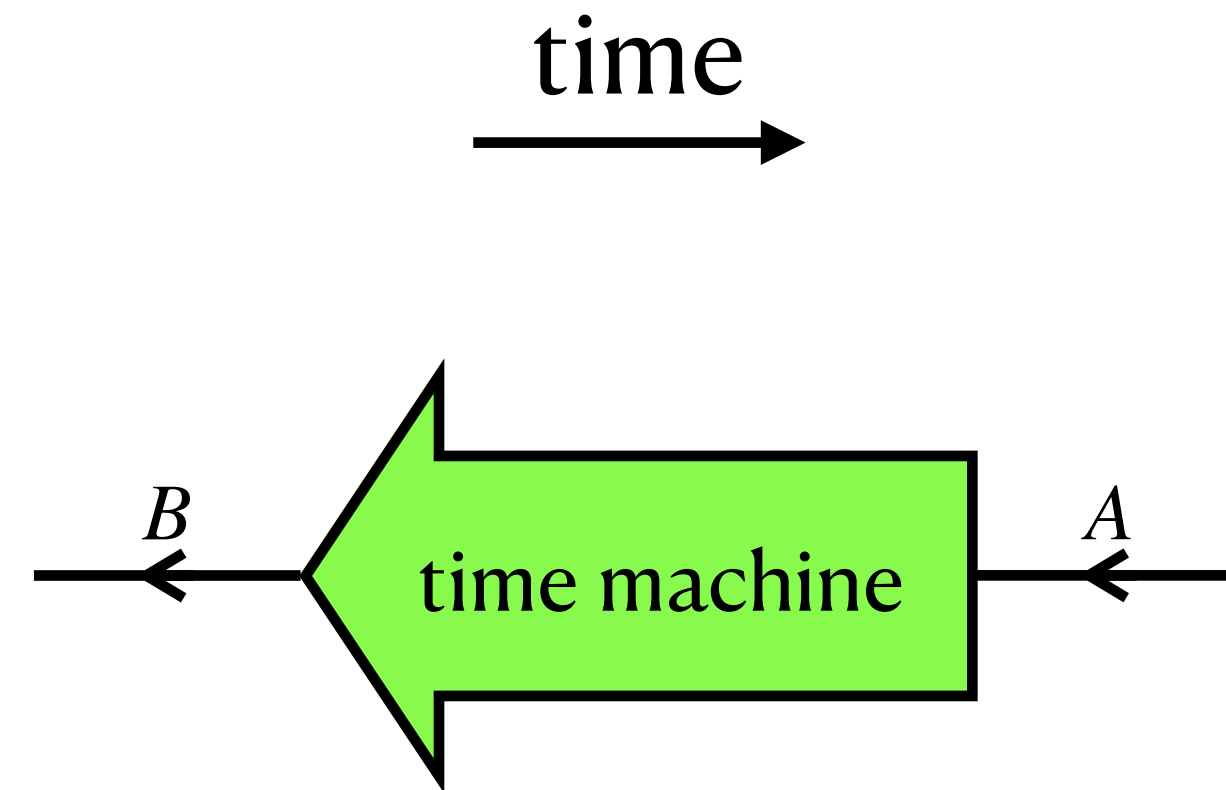
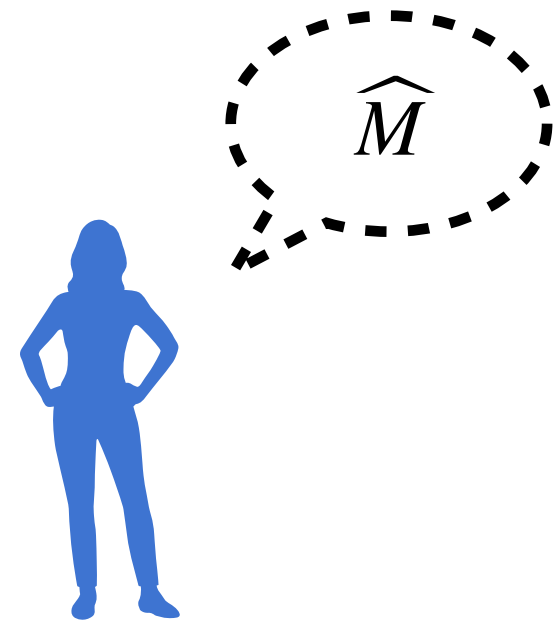


sender (future)

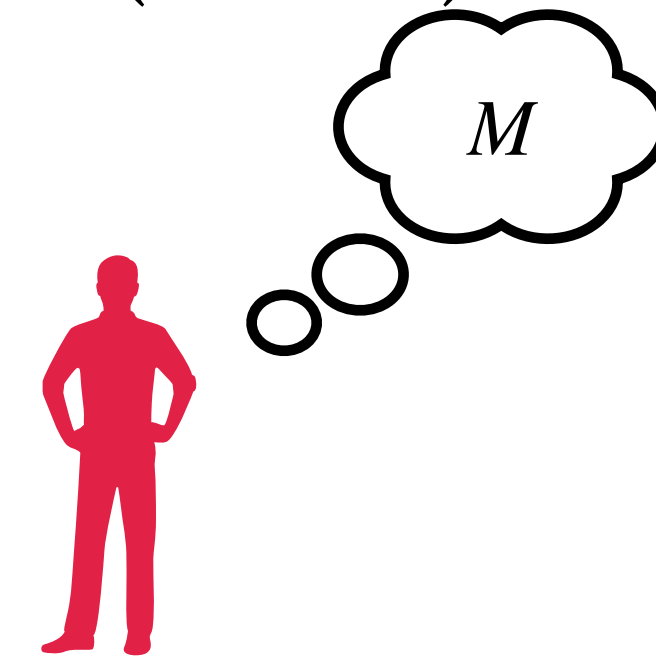


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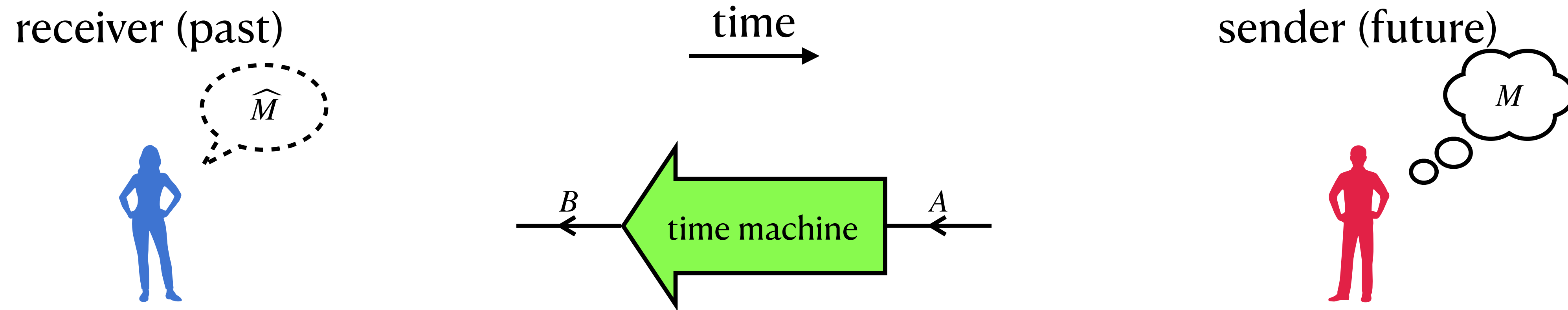


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We study the fundamental limit of communication, backward in time, through a given **noisy time machine** (assuming its existence)

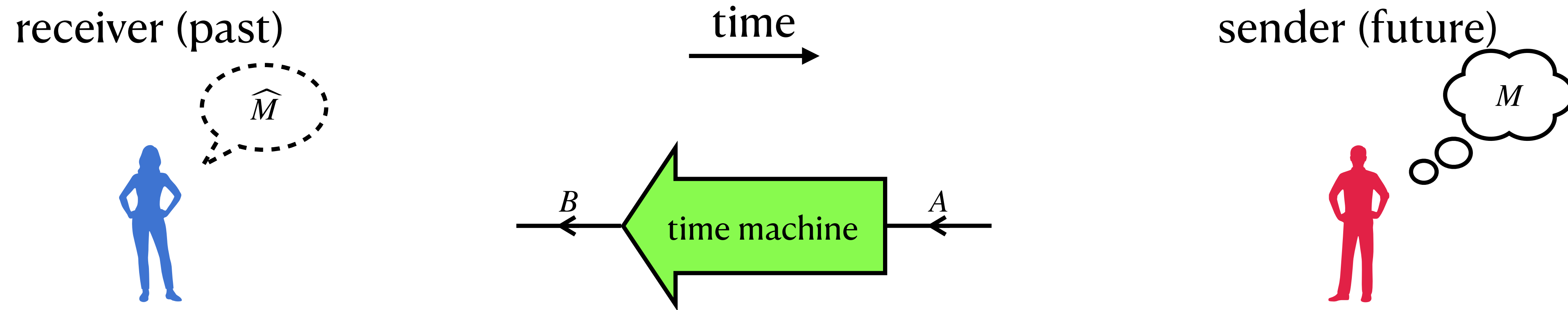
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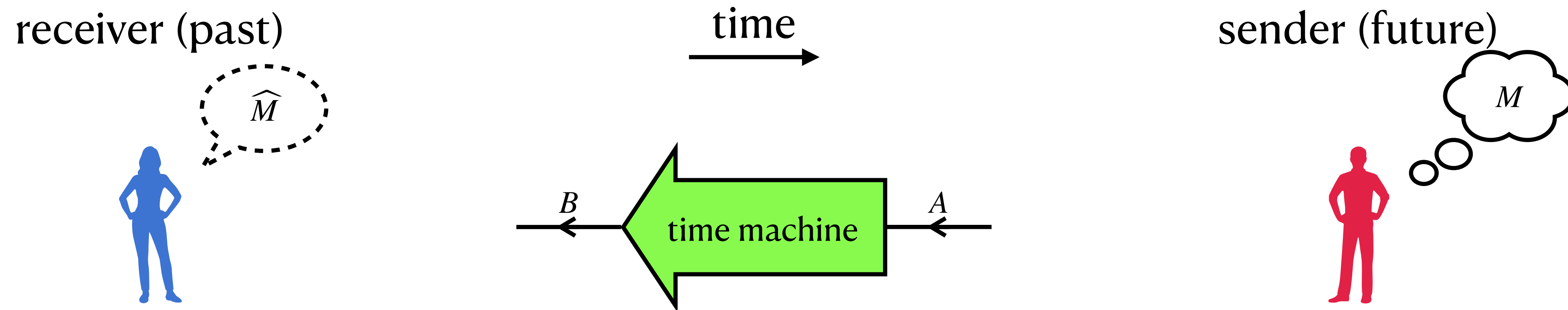


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- “Retrocausal capacity” of $\mathcal{N}_{A \rightarrow B}$

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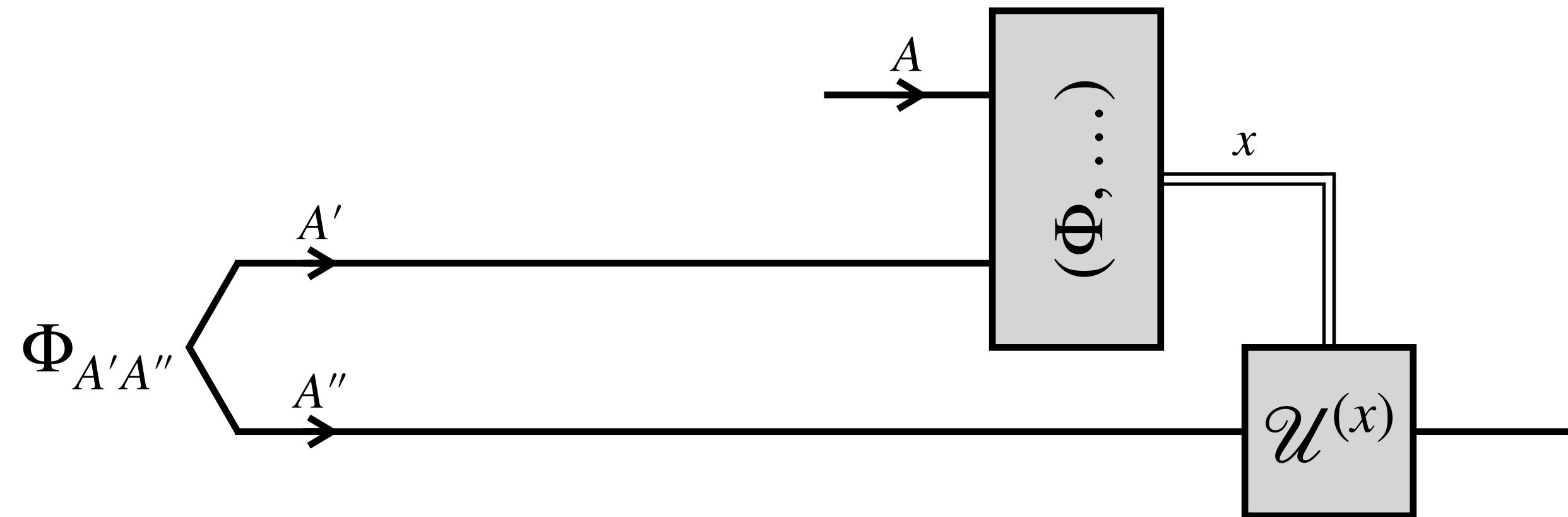
- “Retrocausal capacity” of $\mathcal{N}_{A \rightarrow B}$

We solve this precisely in the one-shot regime

How to describe a noisy time machine mathematically?

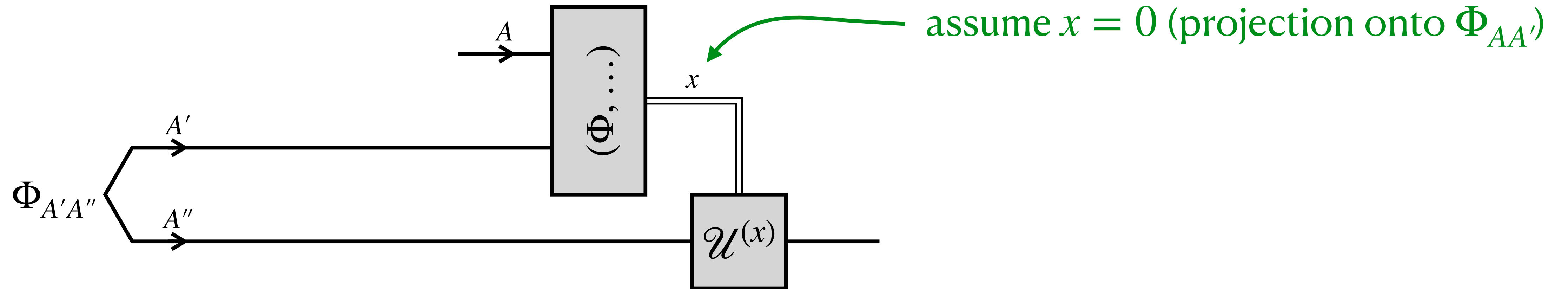
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Teleportation

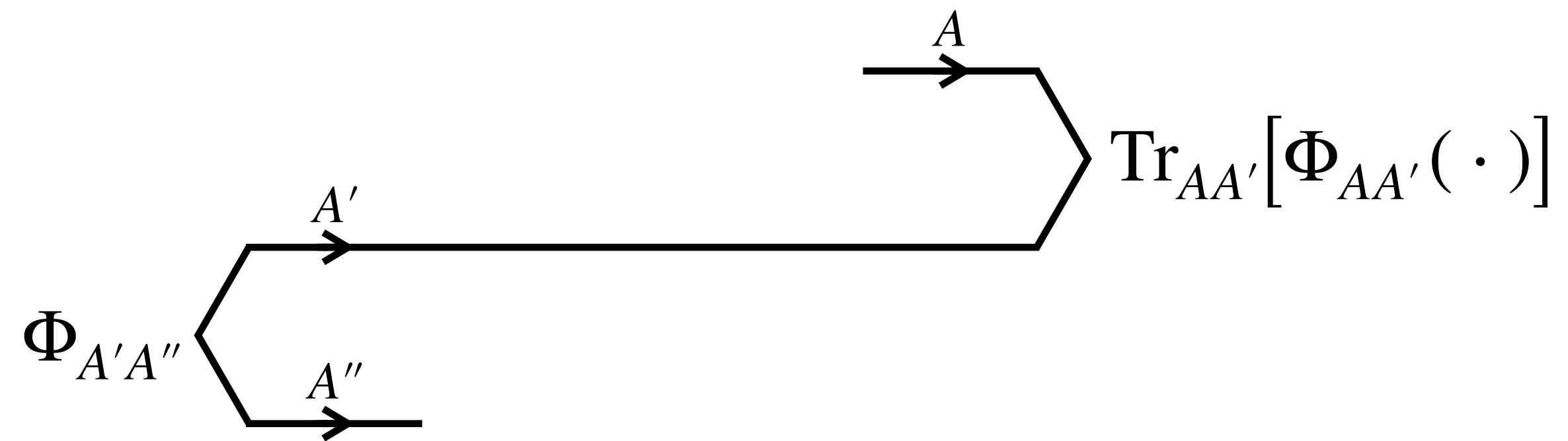


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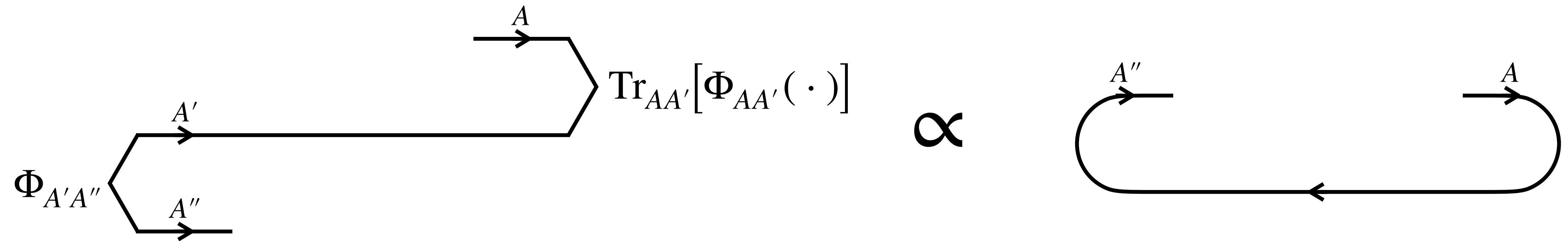
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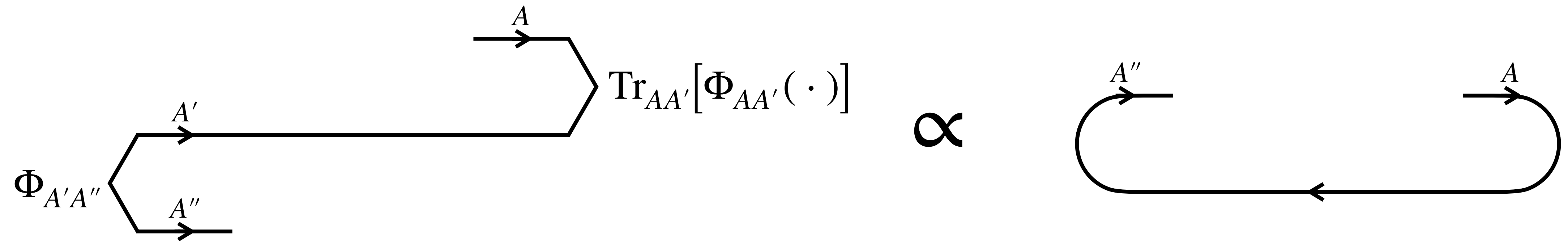


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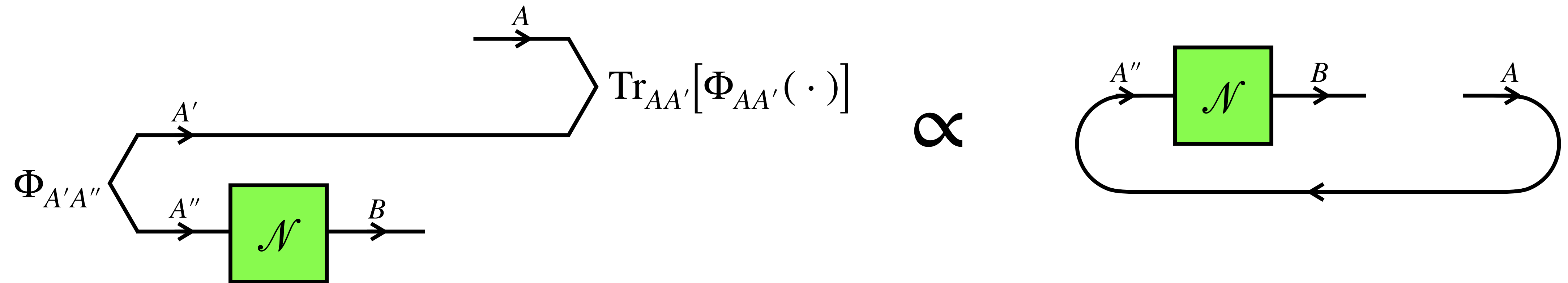
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Postselected closed timelike curve (P-CTC)



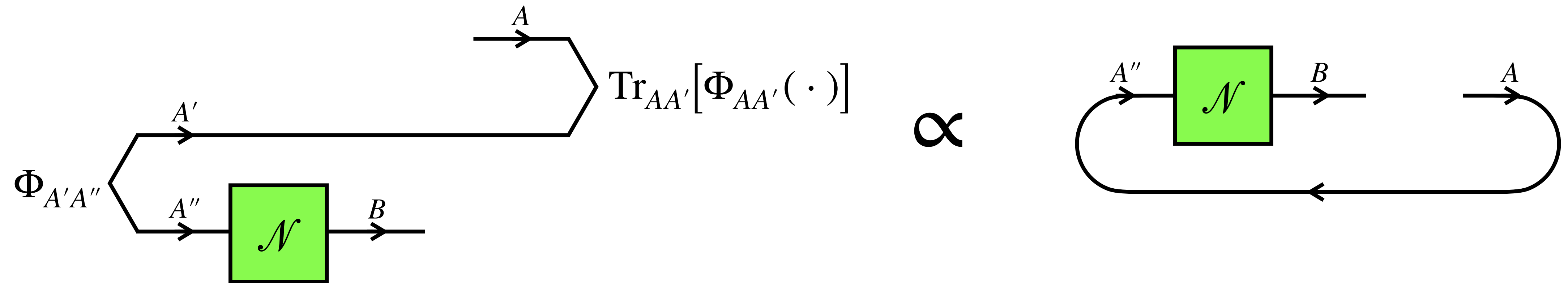
Bennett and Schumacher, unpublished (2002)
 Svetlichny, arXiv:0902.4898 (2009)
 Lloyd et al., PRL 106, 040403 (2011)
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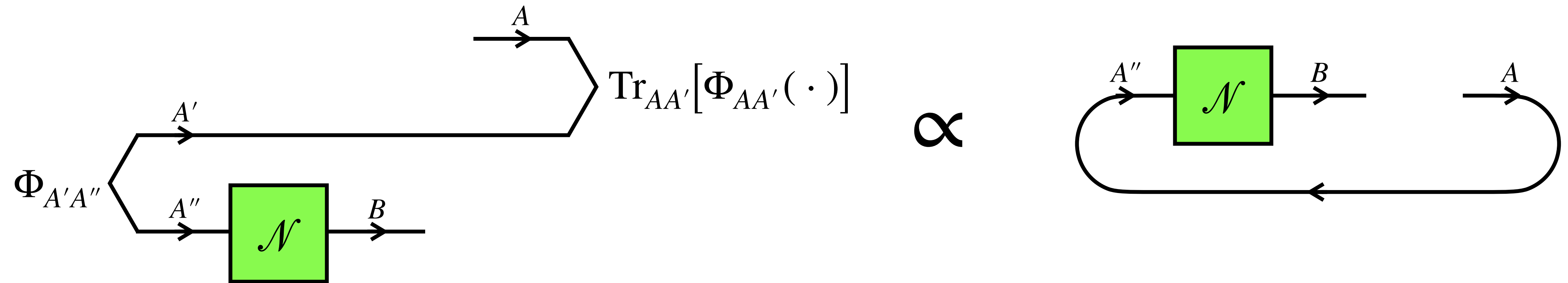
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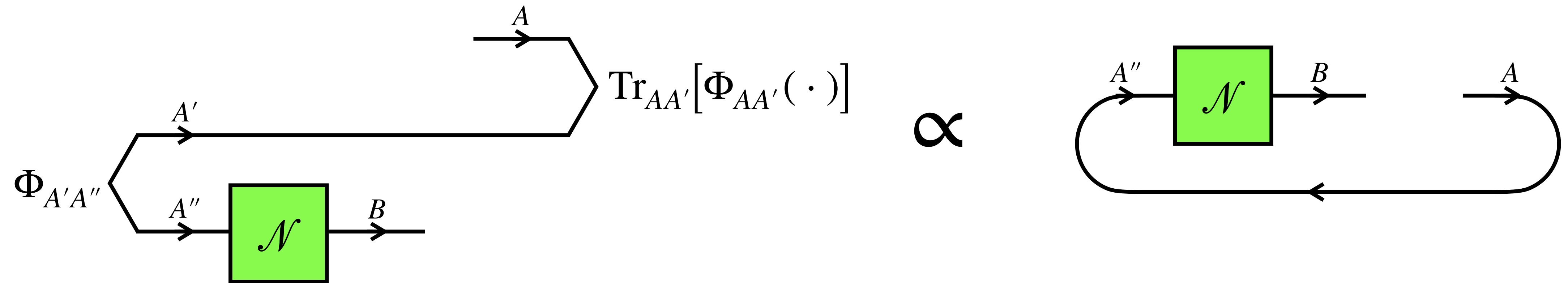
“Noisy” P-CTC



The effect of the noisy P-CTC represented by a CPTP map $\mathcal{N}_{A \rightarrow B}$:

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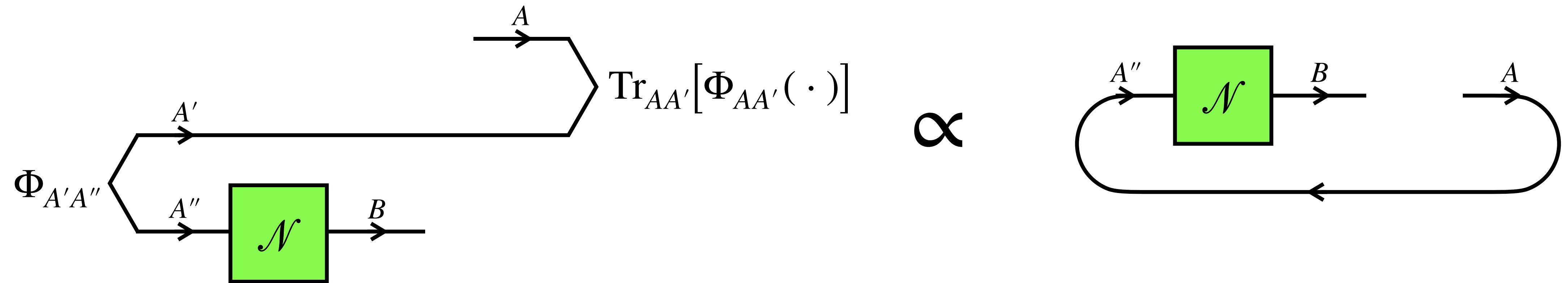


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- Turn a future system A into a past system B

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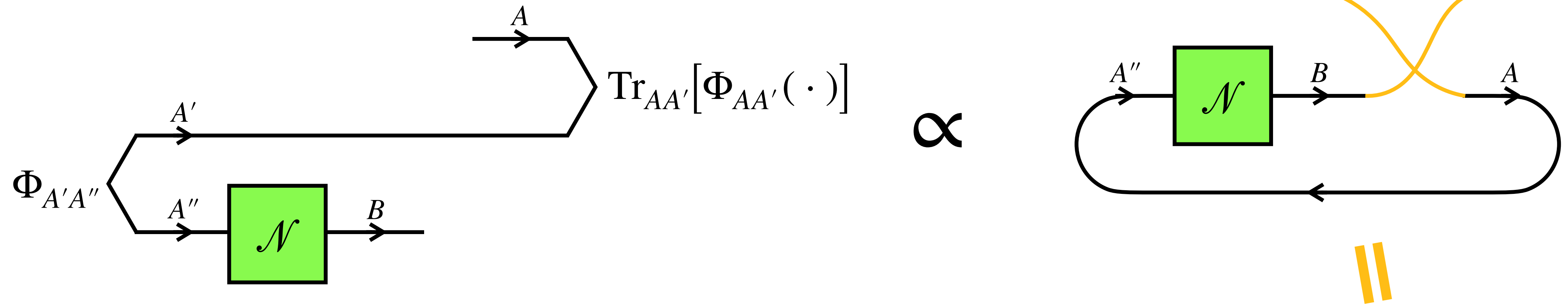


The effect of the noisy P-CTC represented by a CPTP map $\mathcal{N}_{A \rightarrow B}$:

- Turn a future system A into a past system B
- If we wait for B to “catch up”, then it realizes an ordinary channel represented by $\mathcal{N}_{A \rightarrow B}$

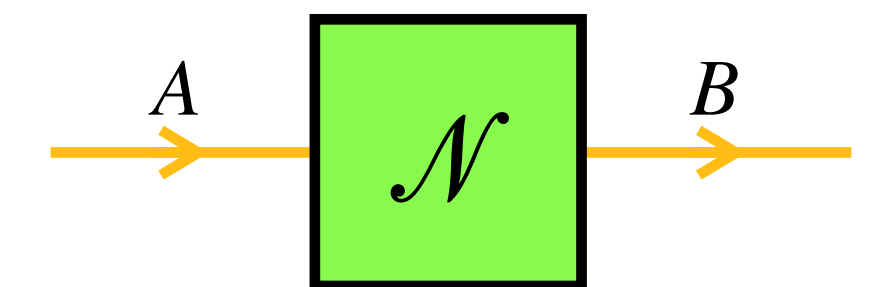
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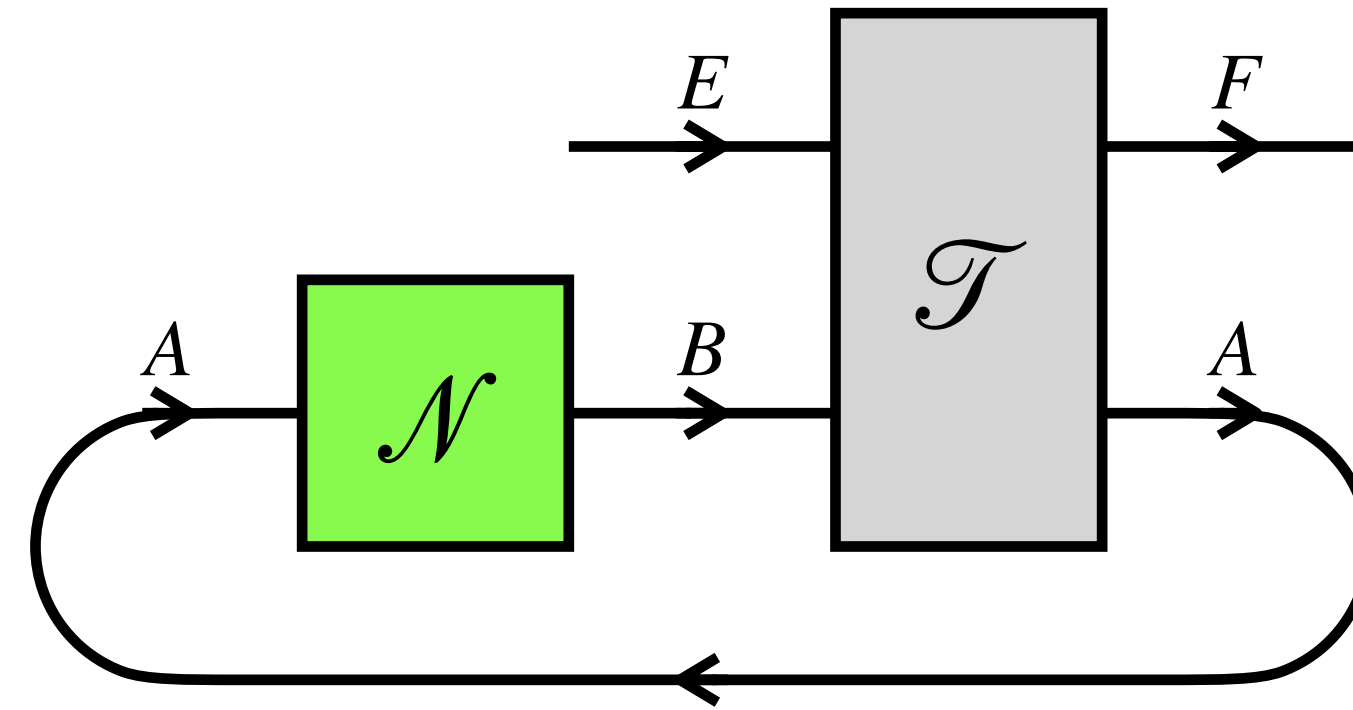


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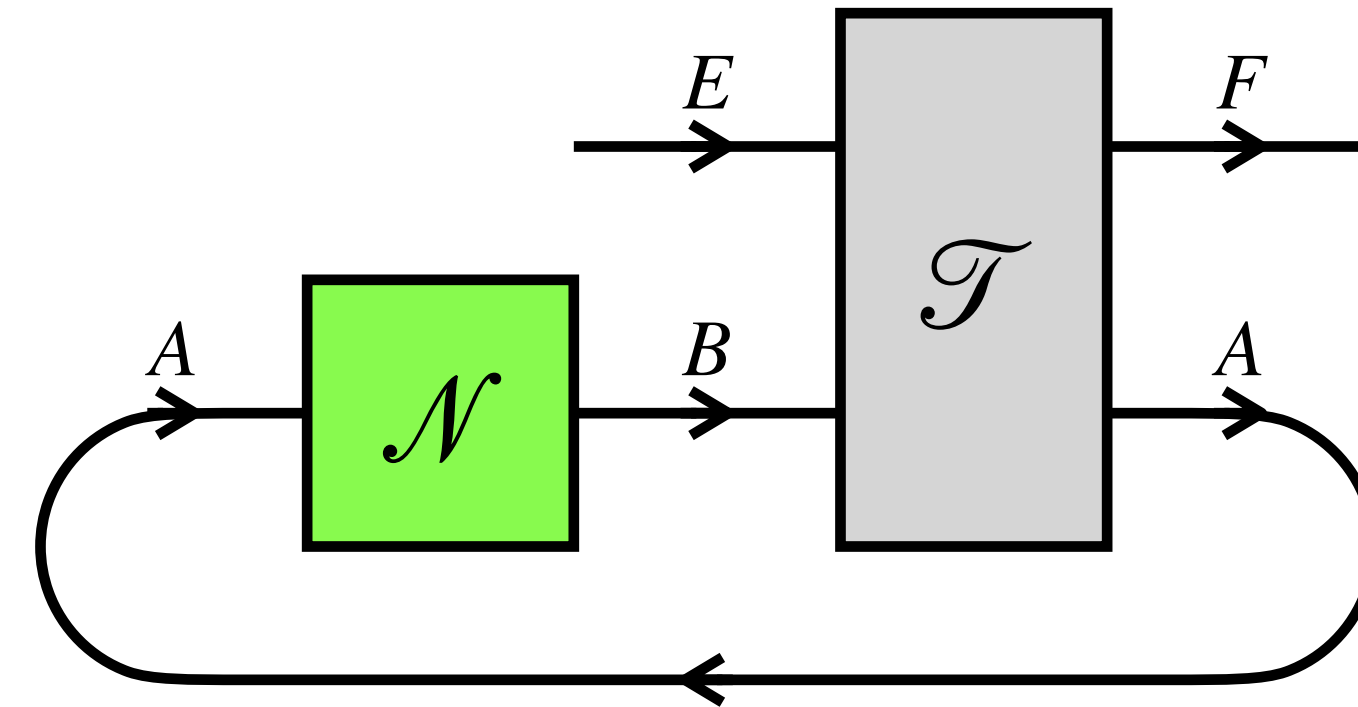


Combine a noisy P-CTC with an ordinary channel



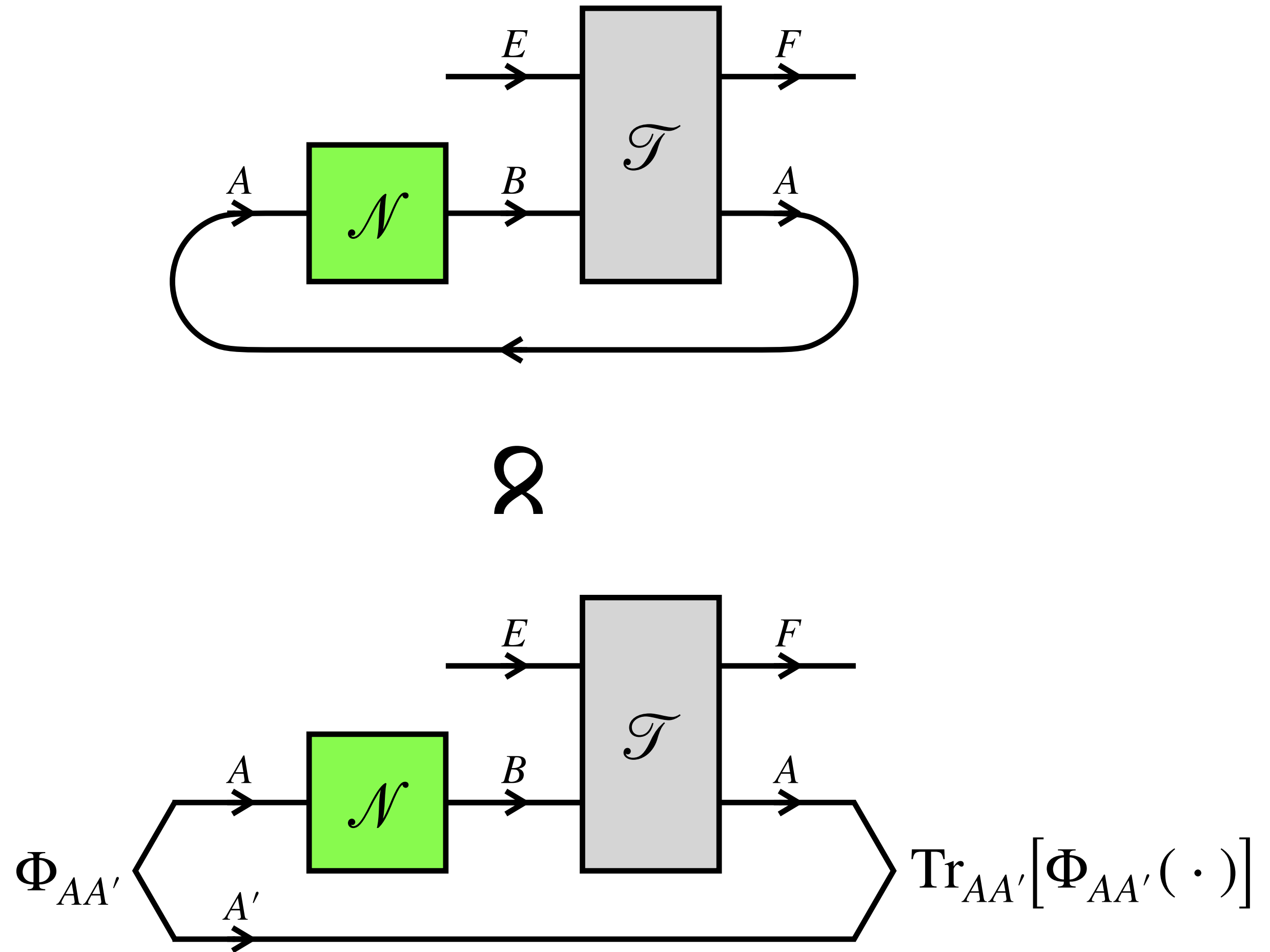
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What is the map from E to F ?



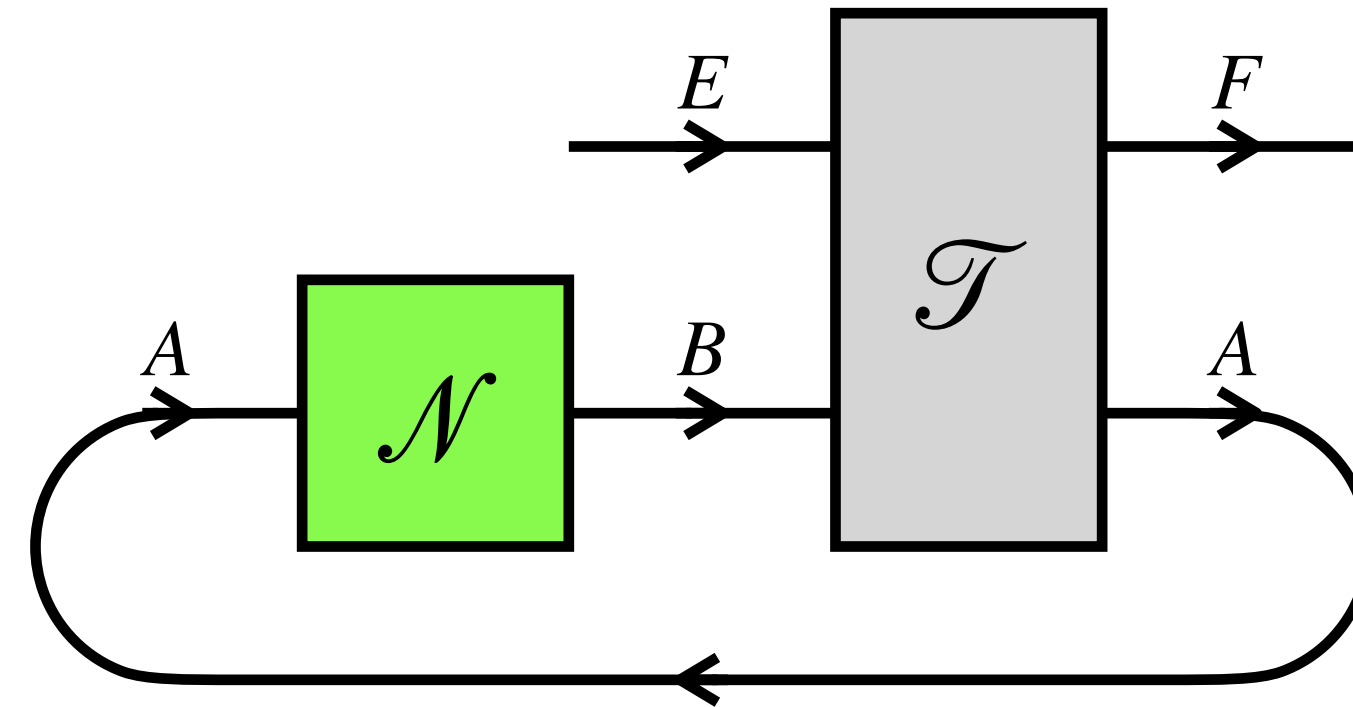
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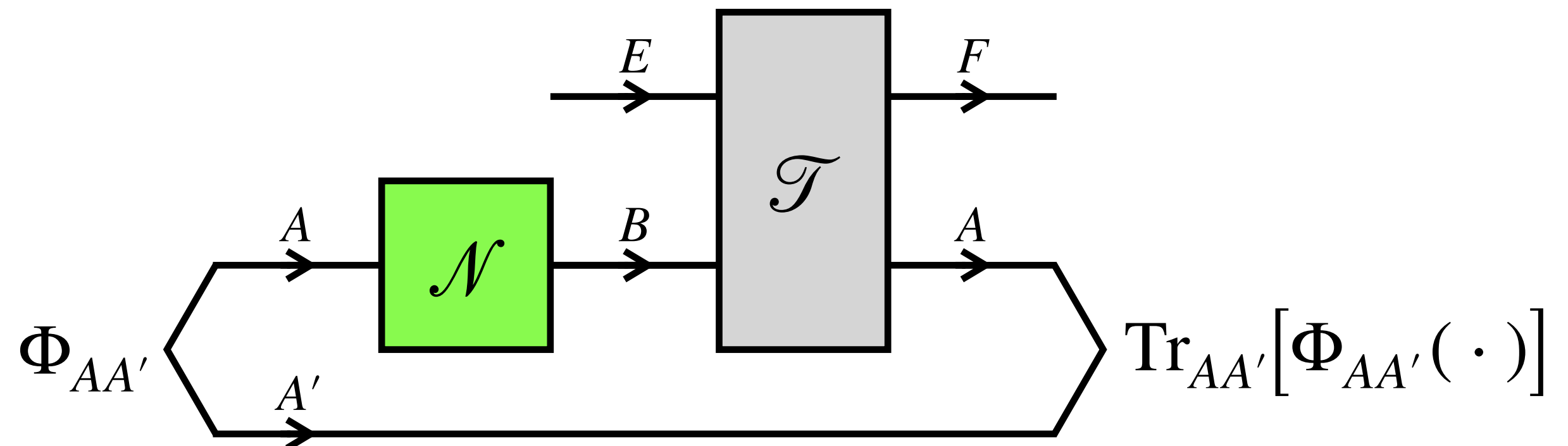
What is the map from E to F ?



$$\rho_{RE} \mapsto \sigma_{RF} \propto \text{Tr}_{AA'}[\Phi_{AA'} \mathcal{T}_{EB \rightarrow FA}[\rho_{RE} \otimes \mathcal{N}_{A \rightarrow B}[\Phi_{AA'}]]]$$

$\bowtie$

- nonlinear due to renormalization



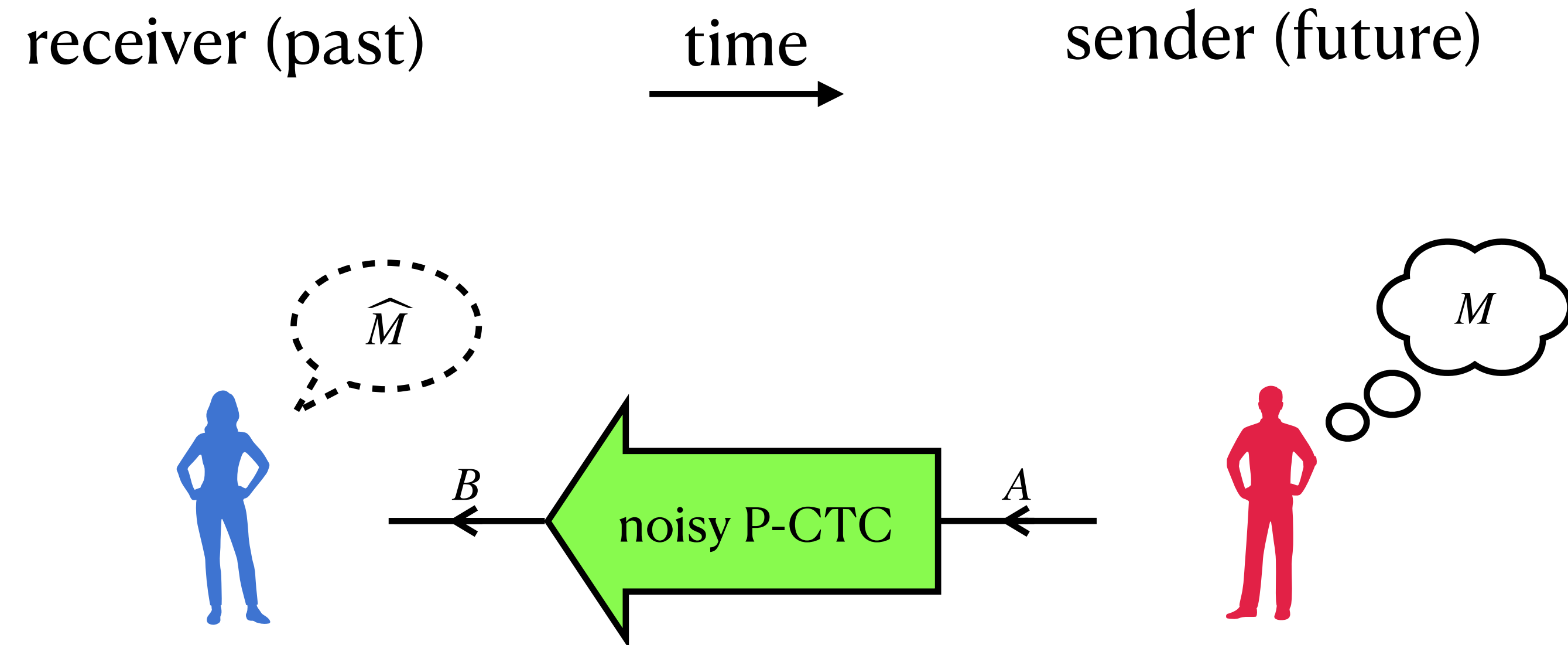
“Retrocausal” communication:

Messaging the past from the future

Retrocausal communication

Setting: a sender in the future is connected to a receiver in the past through a noisy P-CTC

Goal: transmit quantum or classical messages reliably from the sender to the receiver, as efficiently as possible, using whatever strategy allowed by physics

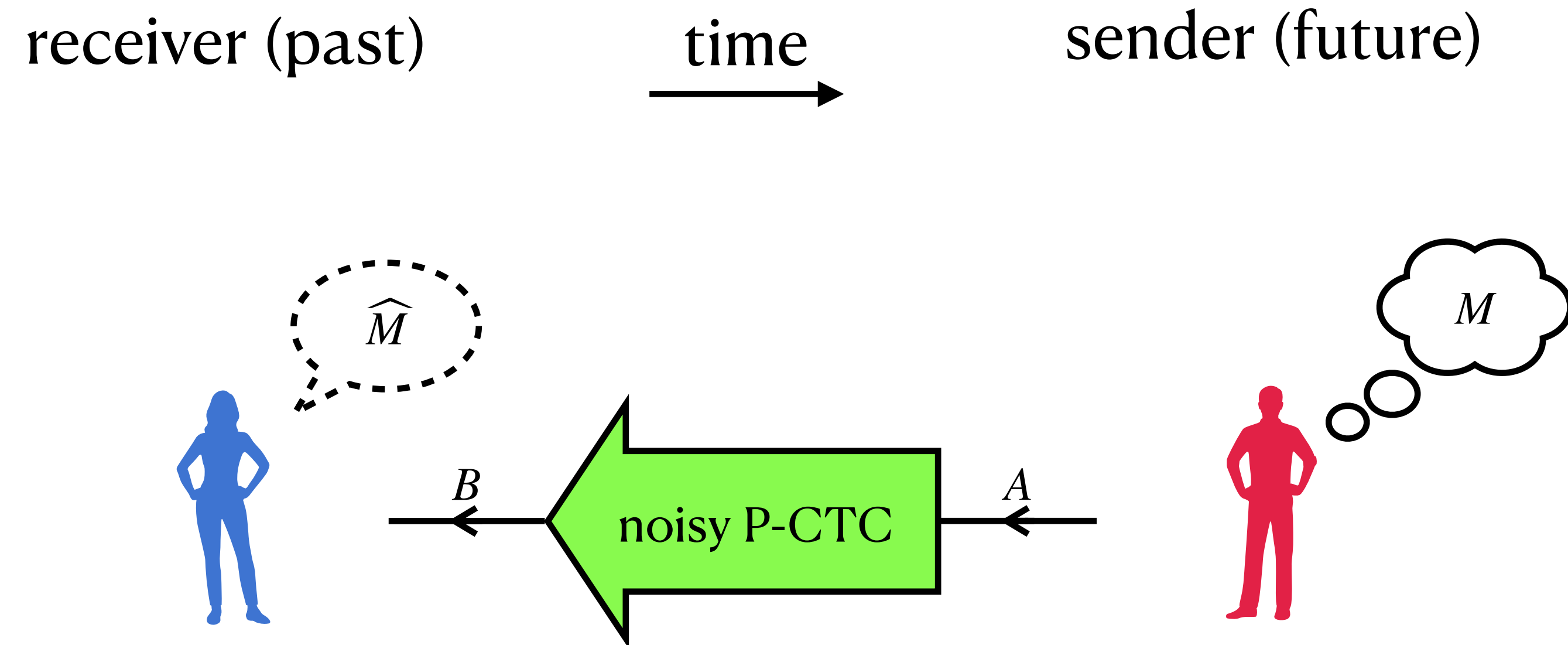


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Isn't this exactly what was happening in the film *Interstellar*?



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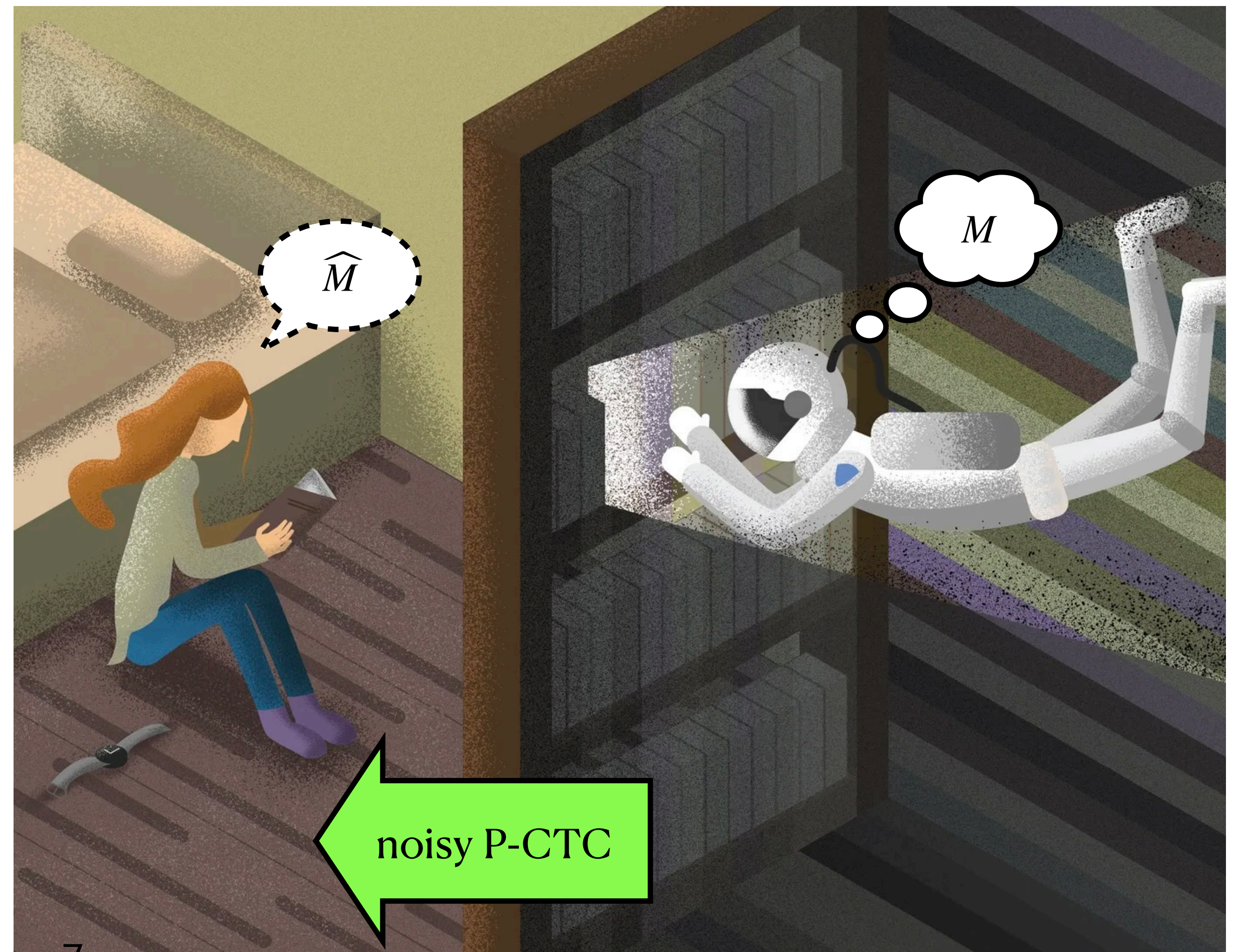
bookshelf

daughter

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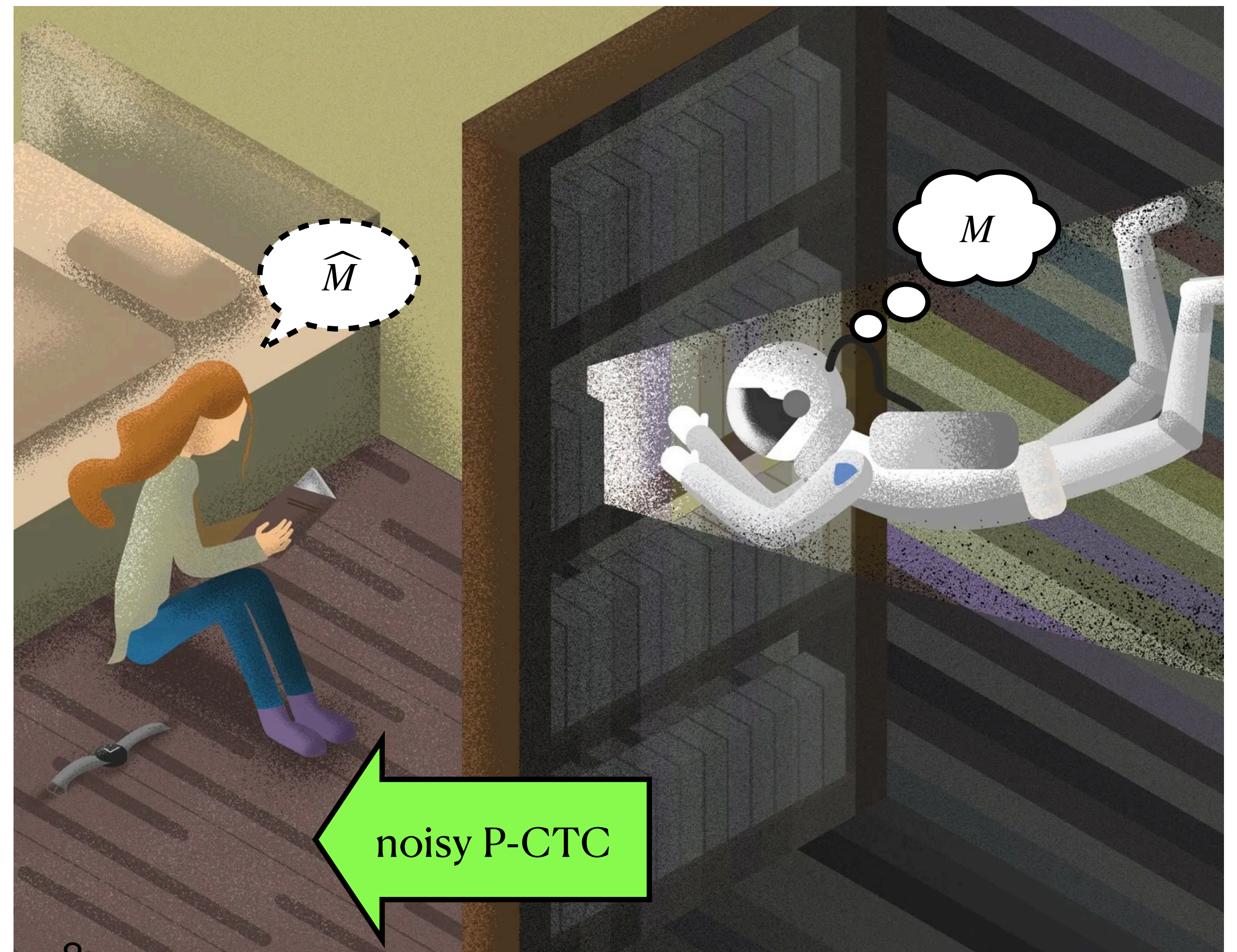
receiver (past) $\xrightarrow{\text{time}}$ sender (future)



receiver (past)

time
→

sender (future)



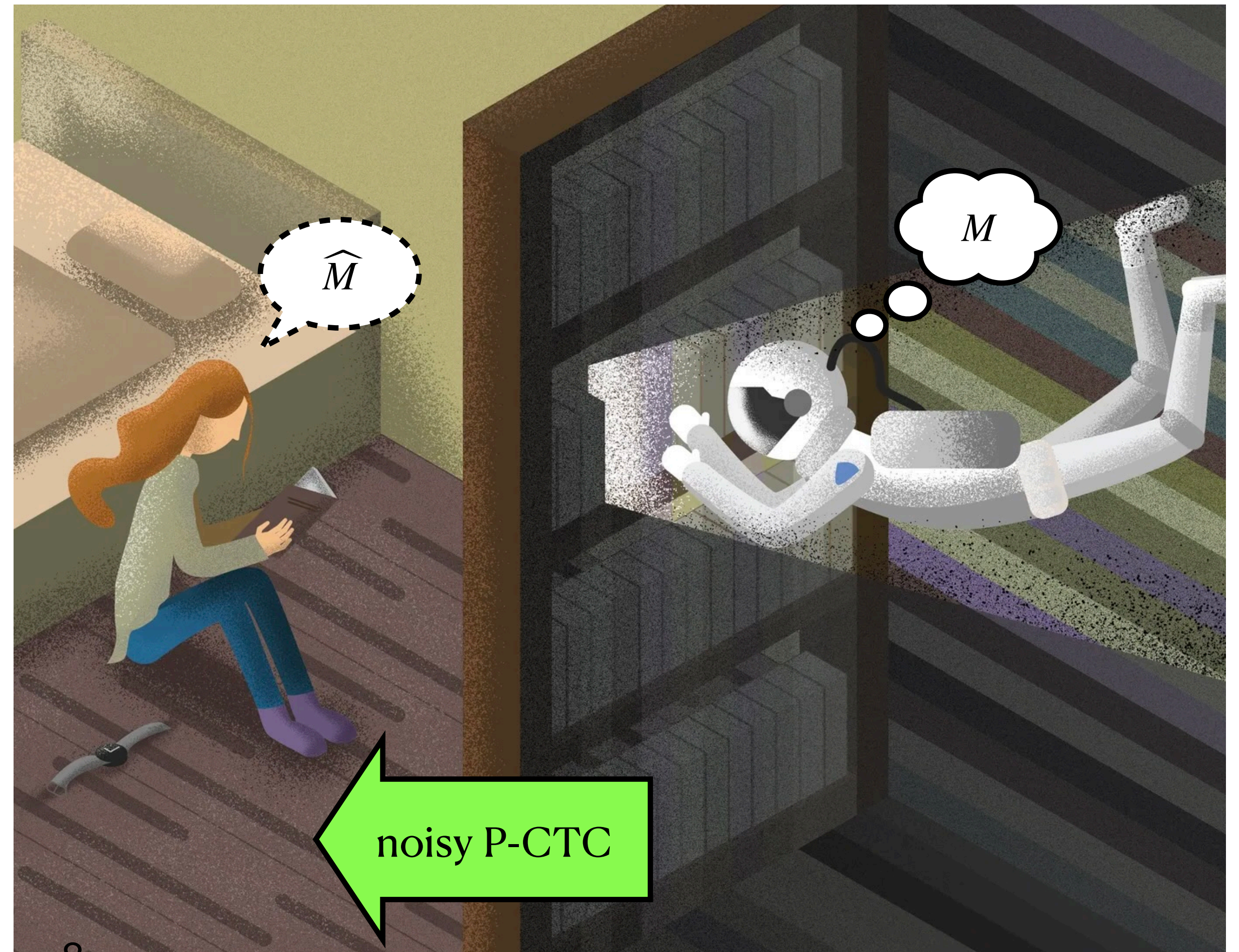
General strategy for retrocausal communication

Strategy:

receiver (past)

time

sender (future)

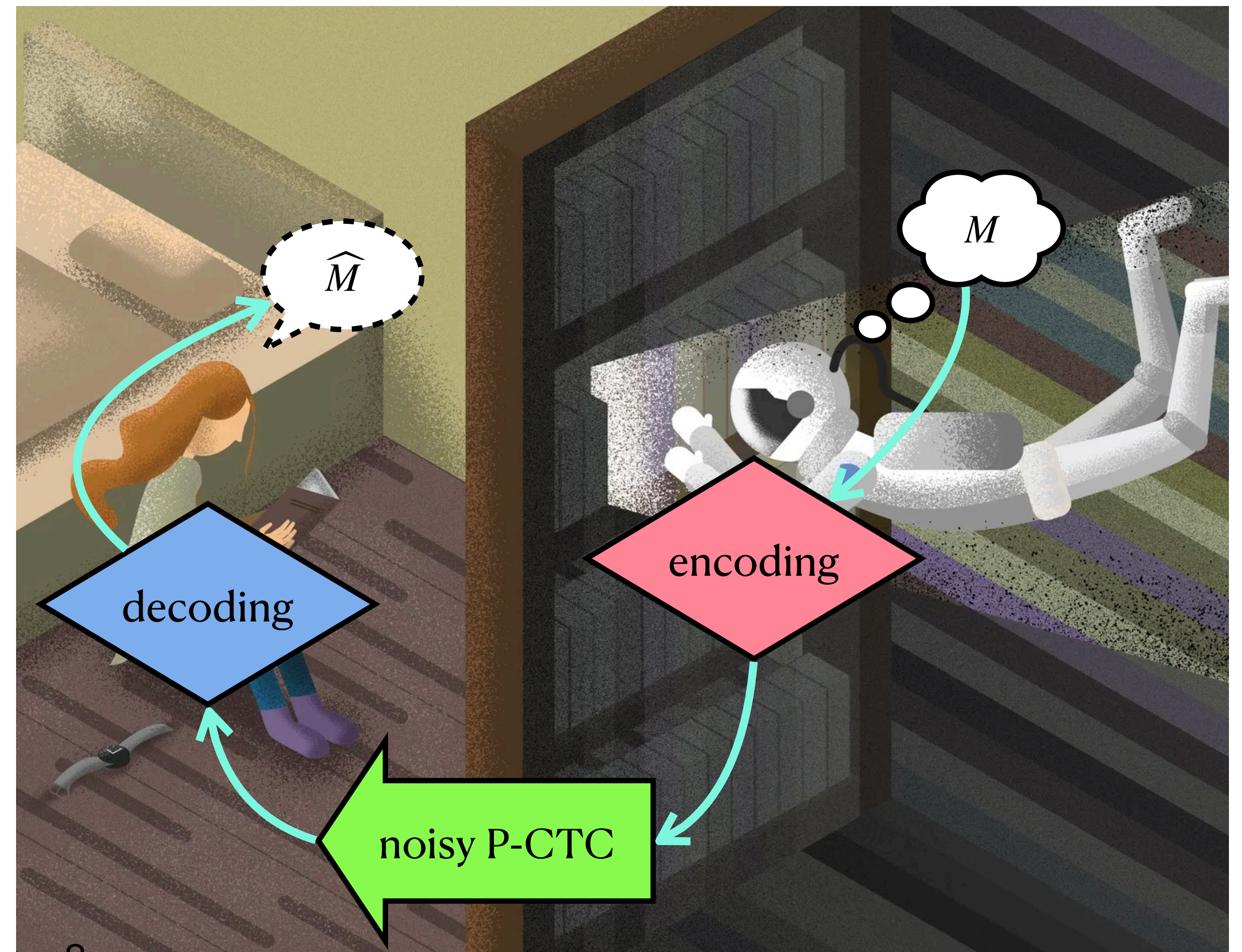


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Strategy:

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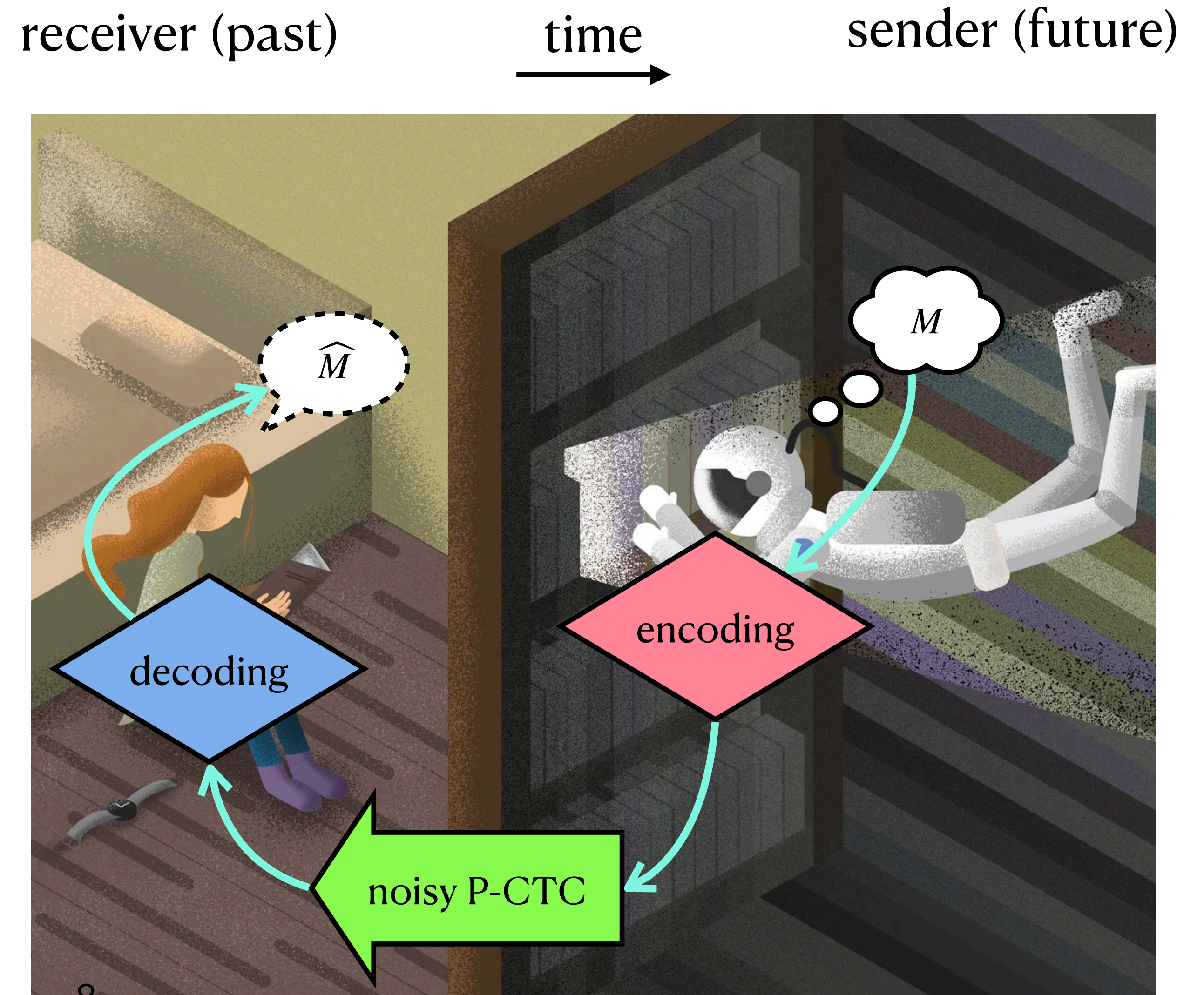


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The encoding, which happens in the future, can be influenced by the decoding, which happens in the past!

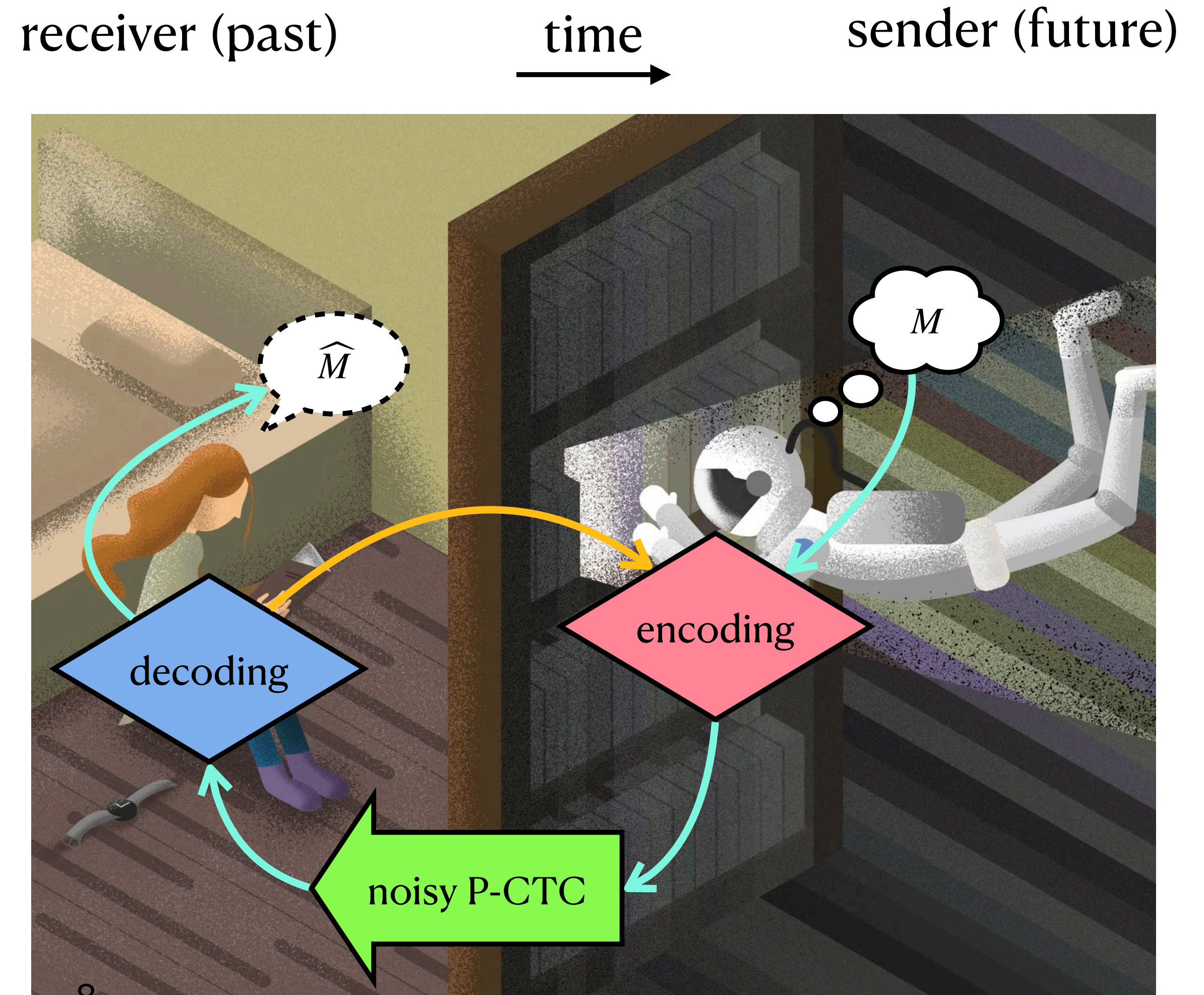


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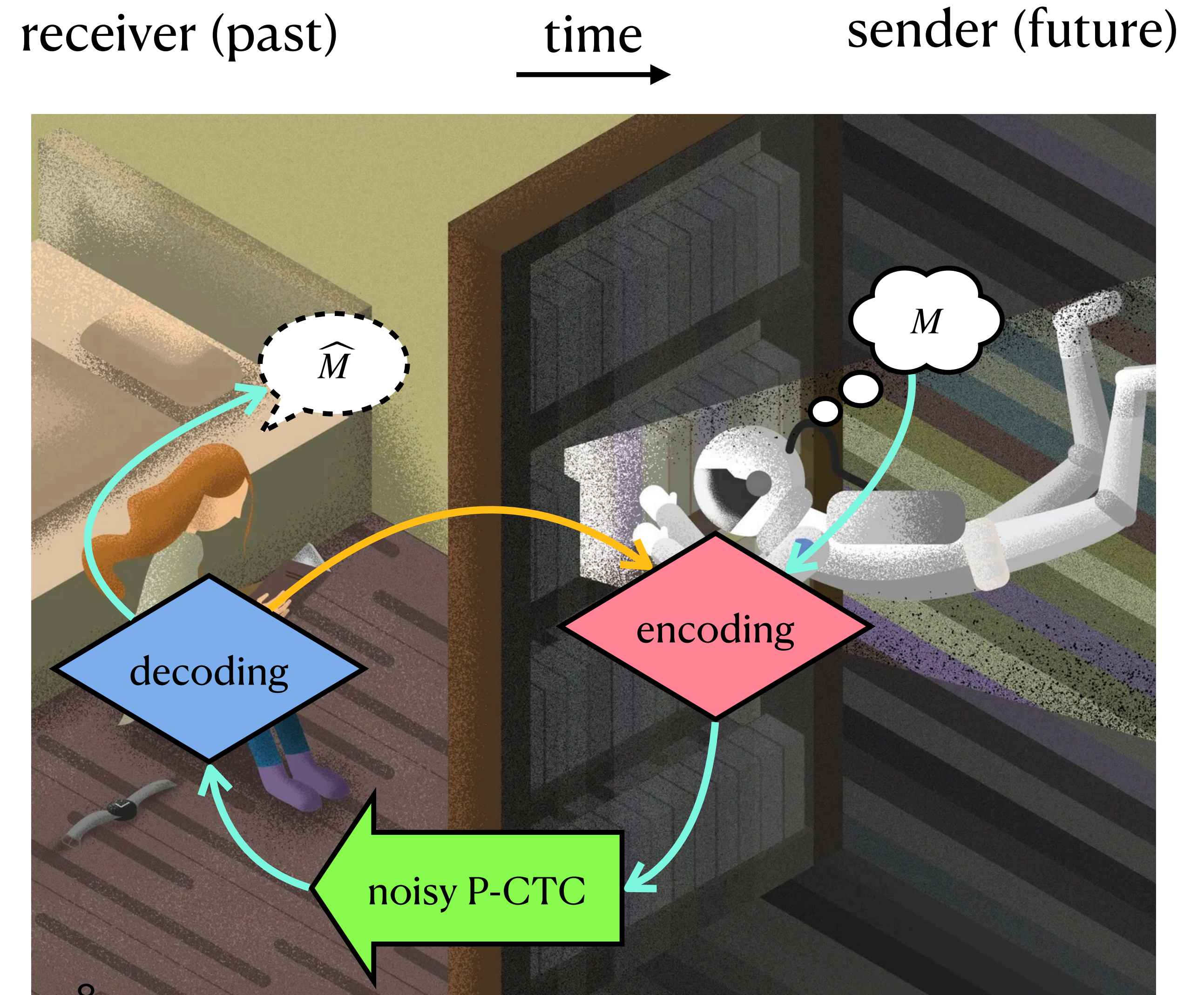
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- Causal loop, but perfectly reasonable



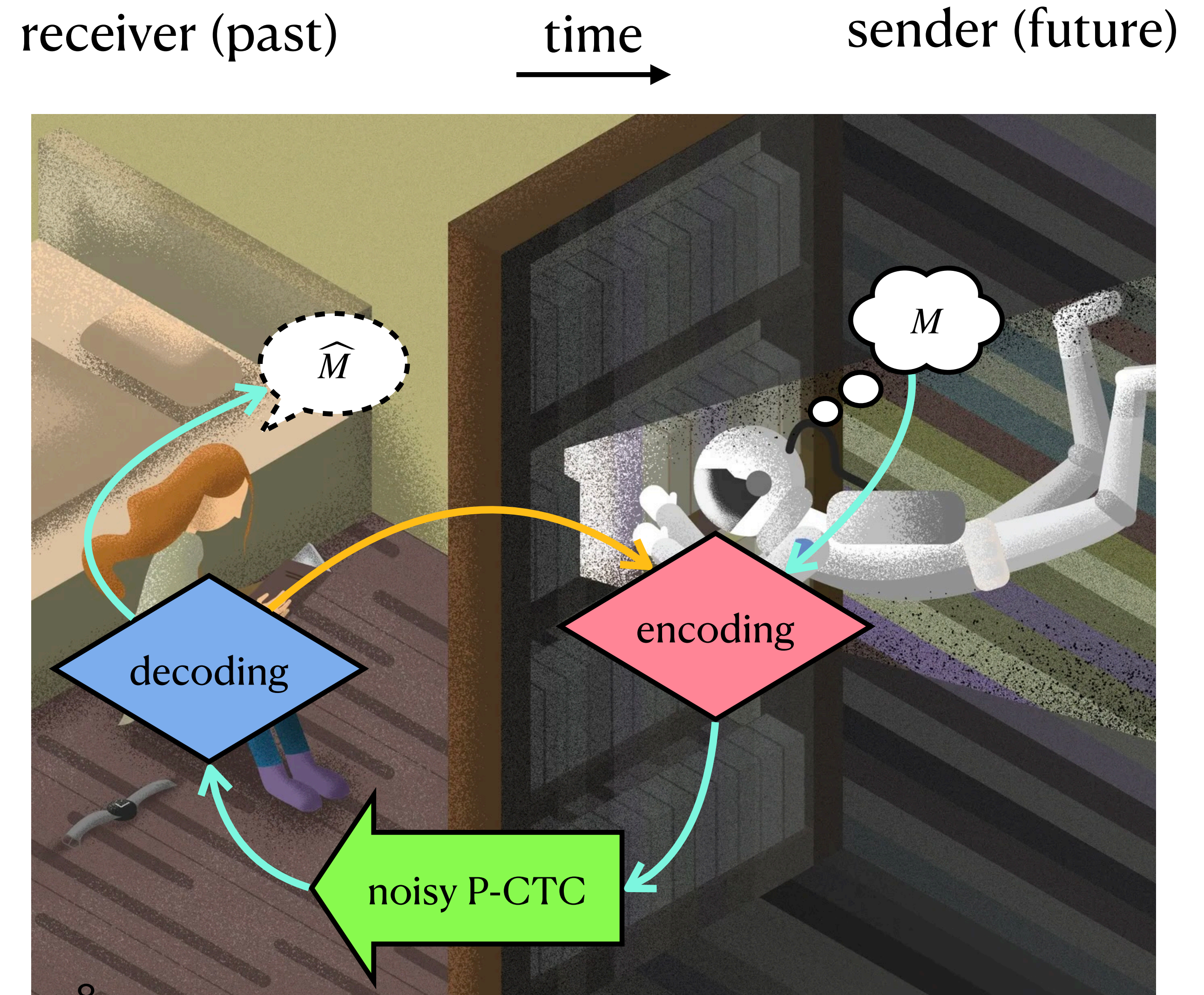
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- Causal loop, but perfectly reasonable
- More powerful than entanglement assistance & nonsignaling assistance



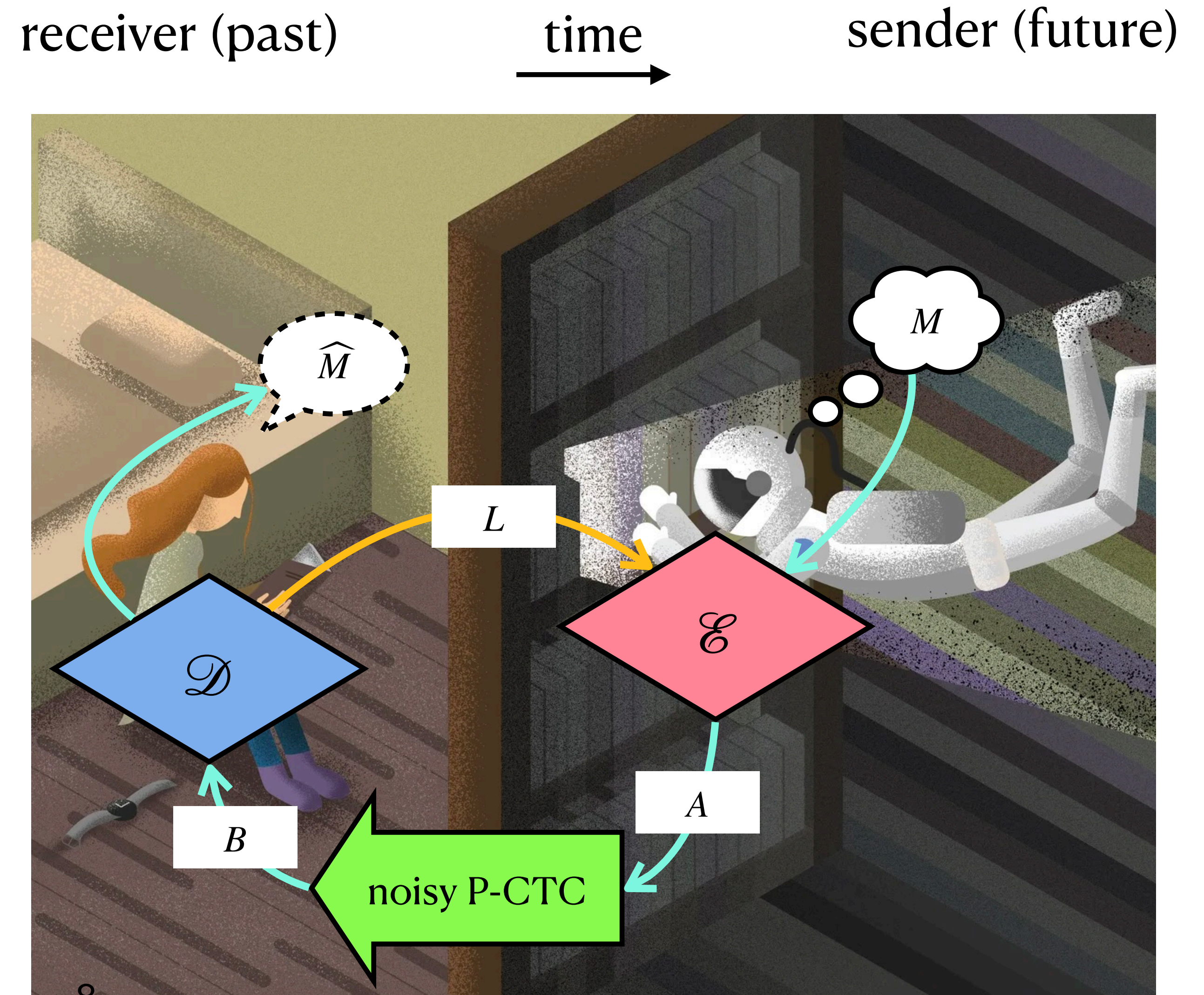
General strategy for retrocausal communication

Strategy:

- Encoding $\mathcal{E}_{ML \rightarrow A}$
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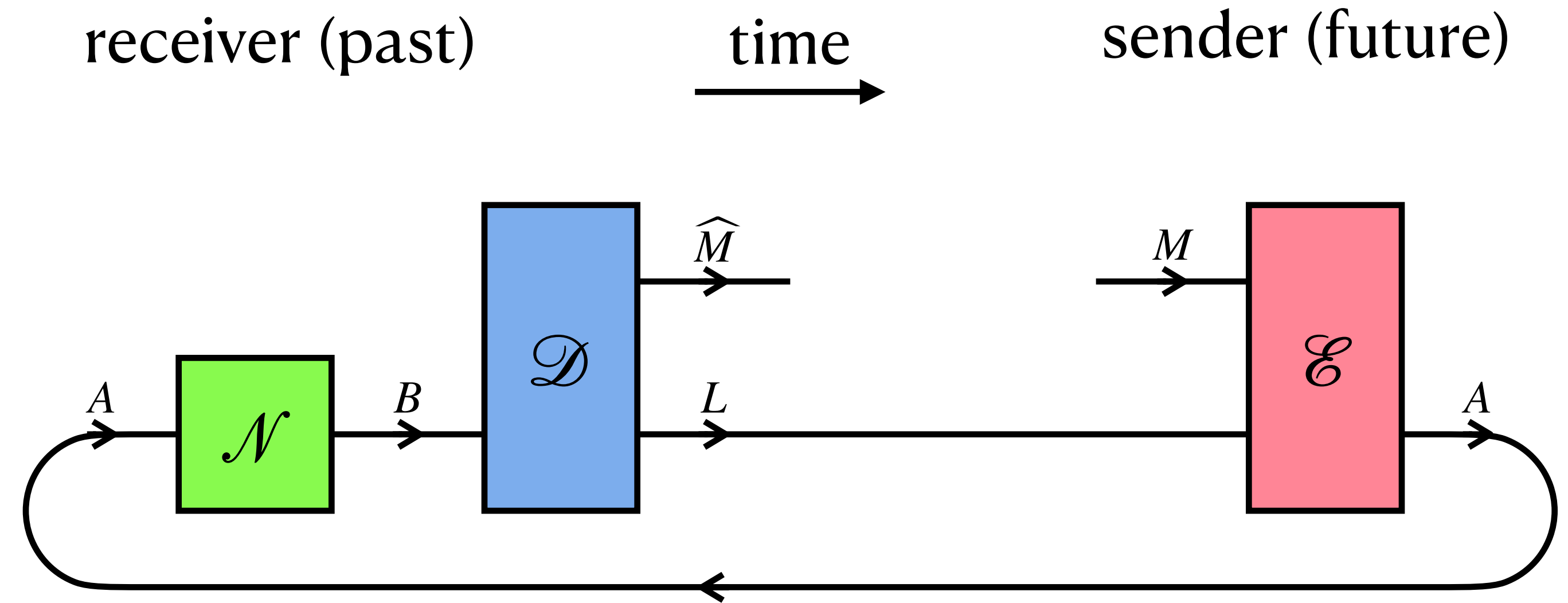
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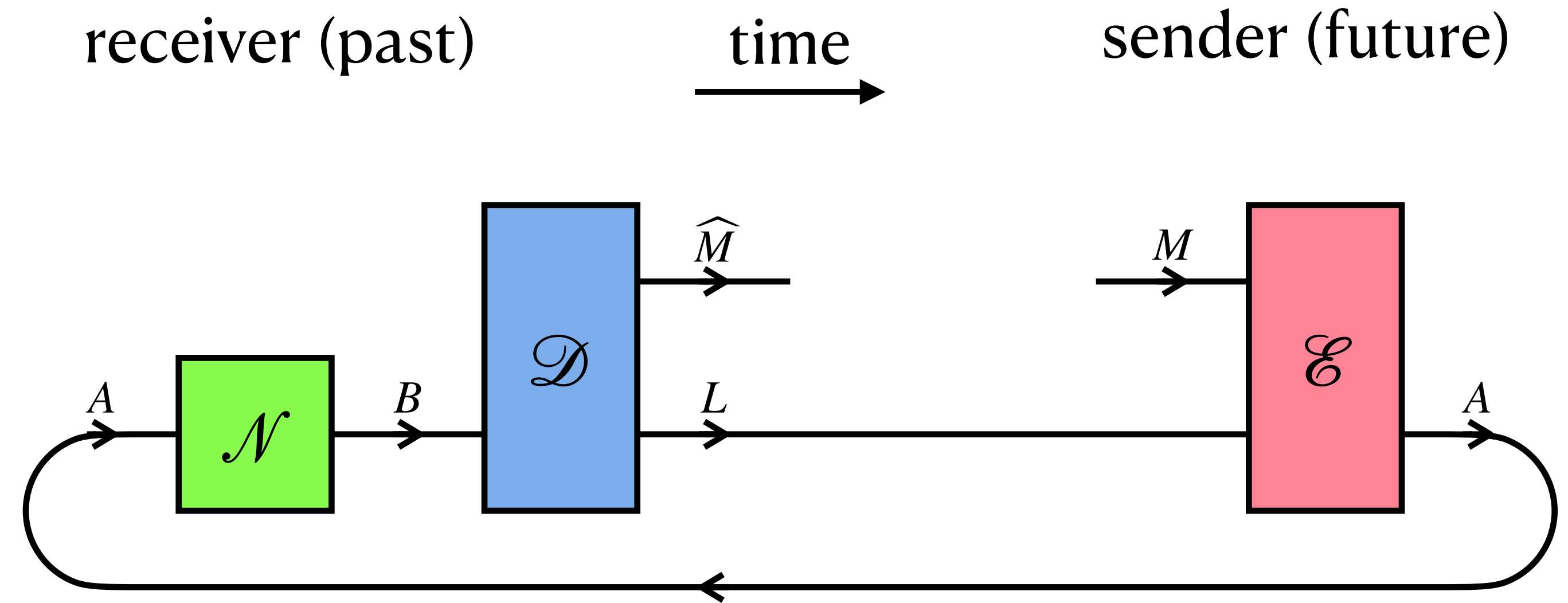
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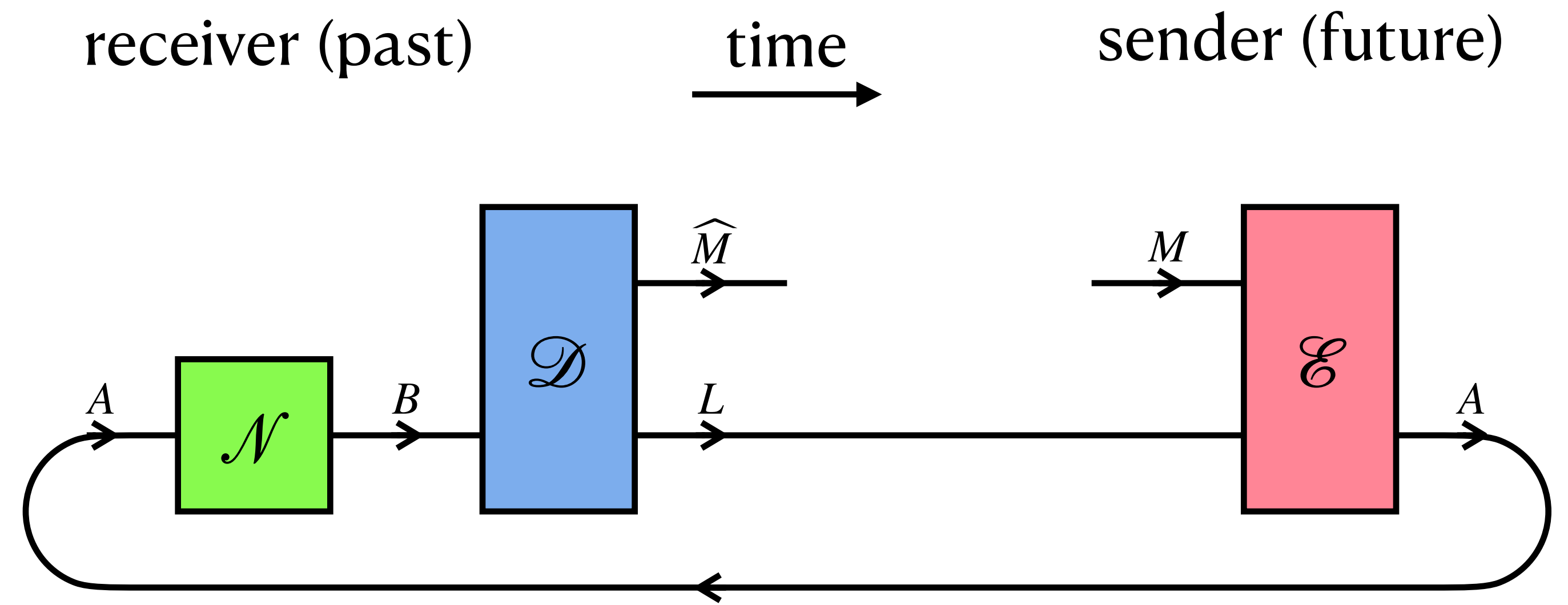
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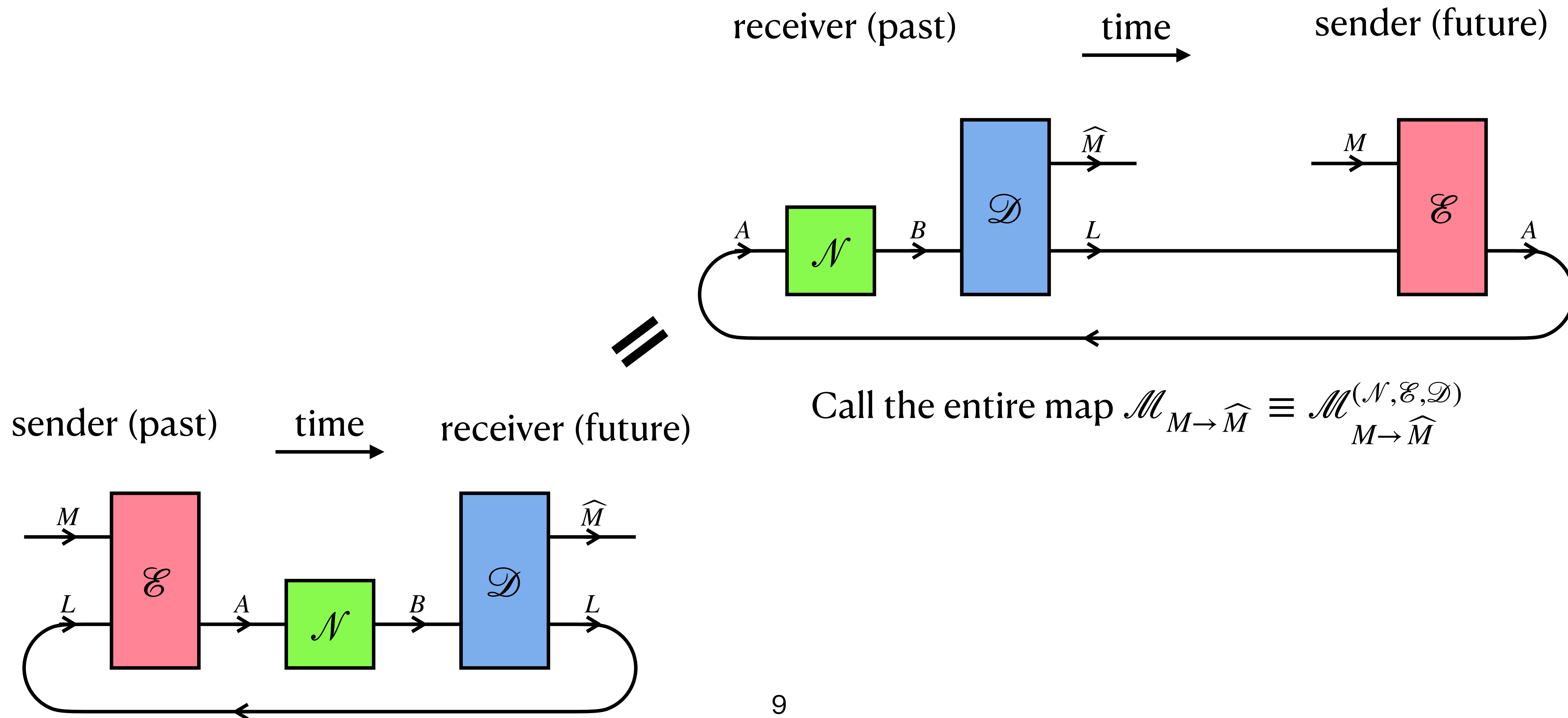
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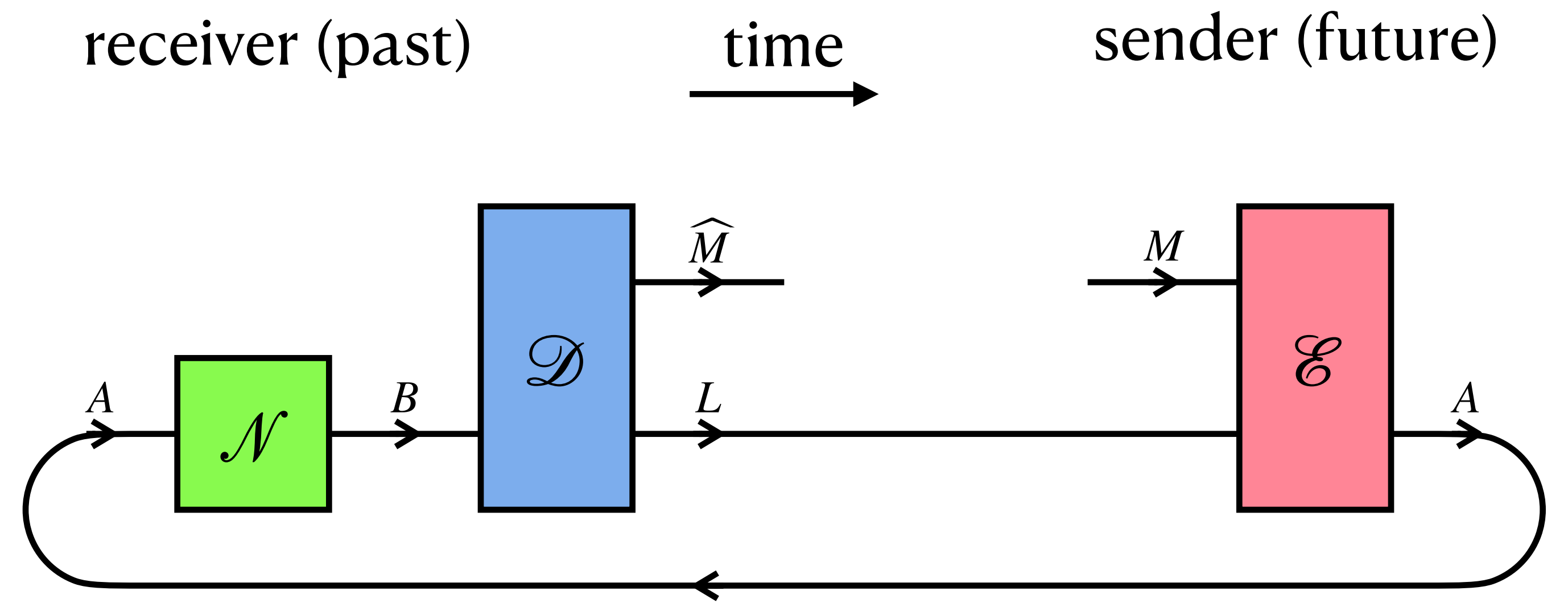


Call the entire map $\mathcal{M}_{M \rightarrow \hat{M}} \equiv \mathcal{M}_{M \rightarrow \hat{M}}^{(\mathcal{N}, \mathcal{E}, \mathcal{D})}$



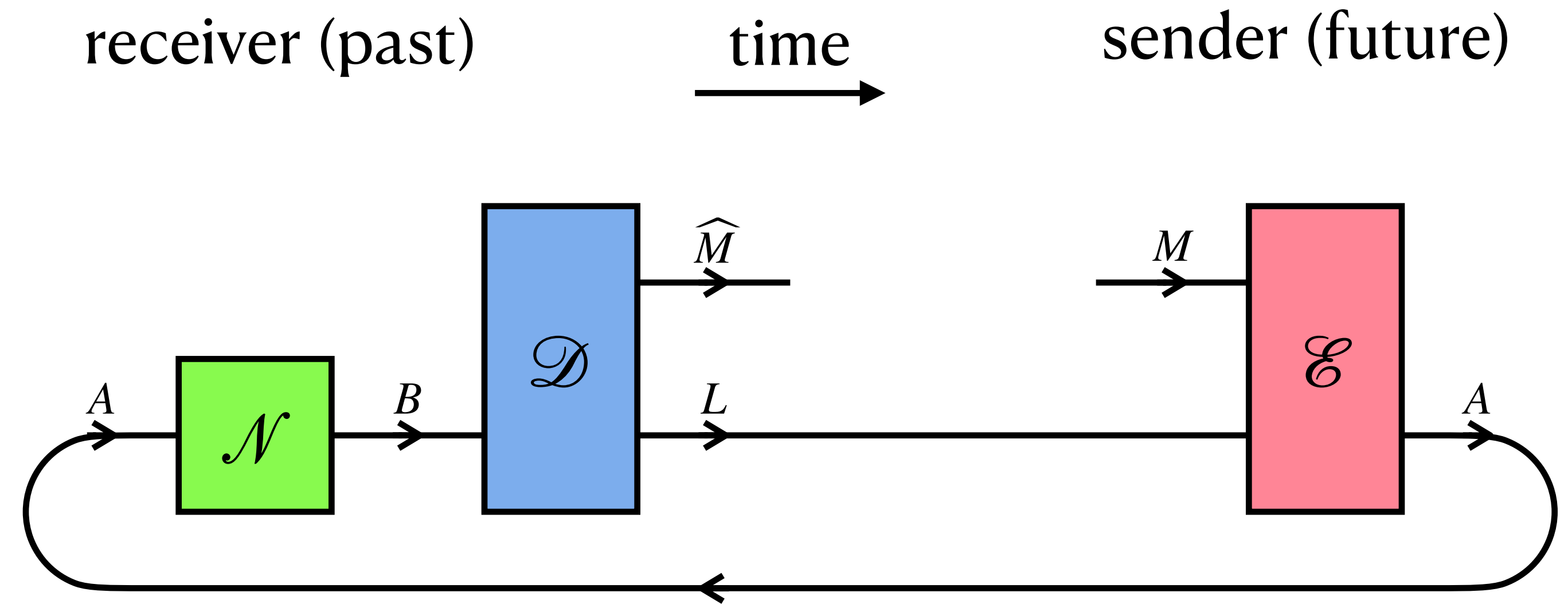
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Definition: One-shot retrocausal quantum & classical capacity



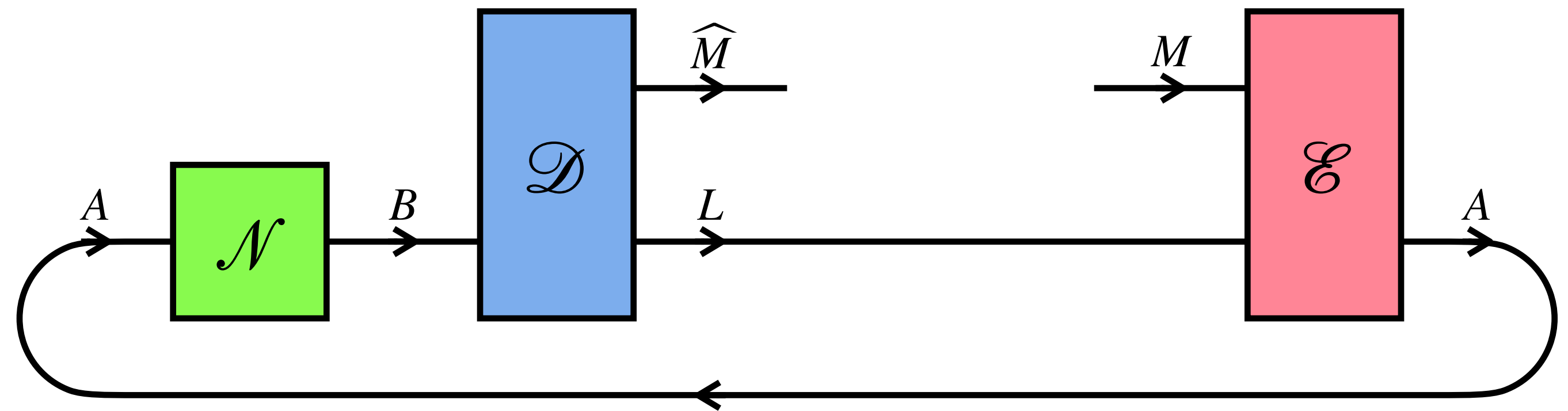
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Definition: One-shot retrocausal quantum & classical capacity

One-shot retrocausal quantum capacity: receiver (past) $\xrightarrow{\text{time}}$ sender (future)

$$Q_{\text{retro}}^{\varepsilon}(\mathcal{N}) := \sup_{\mathcal{E}_{ML \rightarrow A}, \mathcal{D}_{B \rightarrow \hat{M}L}} \log_2 d_M$$

$$\text{s.t. } \inf_{\psi_{RM}} \text{Tr}[\psi_{R\hat{M}} \mathcal{M}_{M \rightarrow \hat{M}}[\psi_{RM}]] \geq 1 - \varepsilon$$



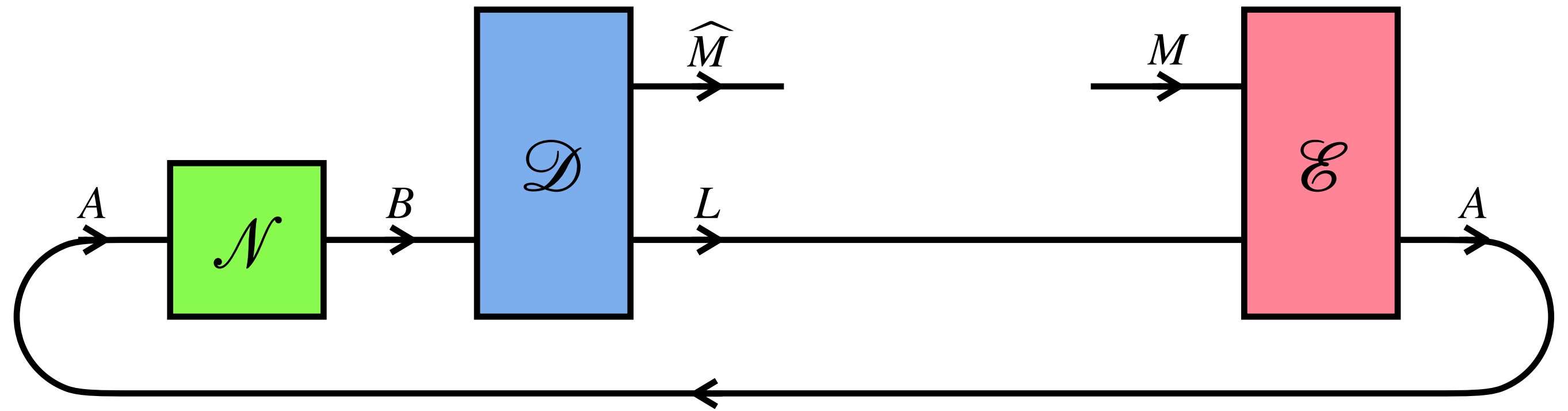
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One-shot retrocausal classical capacity:

$$C_{\text{retro}}^{\varepsilon}(\mathcal{N}) := \sup_{\mathcal{E}_{ML \rightarrow A}, \mathcal{D}_{B \rightarrow \hat{M}L}} \log_2 d_M$$

$$\text{s.t. } \min_{1 \leq m \leq d_M} \text{Tr}[|m\rangle\langle m|_{\hat{M}} \mathcal{M}_{M \rightarrow \hat{M}}[|m\rangle\langle m|_M]] \geq 1 - \varepsilon$$

Call the entire map $\mathcal{M}_{M \rightarrow \hat{M}} \equiv \mathcal{M}_{M \rightarrow \hat{M}}^{(\mathcal{N}, \mathcal{E}, \mathcal{D})}$

Main result: Noisy-“time-machine” coding theorem

One-shot retrocausal quantum & classical capacity: for all $\varepsilon \in [0,1]$,

$$Q_{\text{retro}}^{\varepsilon}(\mathcal{N}) = \log_2 \left[\sqrt{\frac{\varepsilon}{1-\varepsilon} 2^{I_{\text{max}}(\mathcal{N}) + I_{\text{doe}}(\mathcal{N})} + 1} \right]$$

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Asymptotic retrocausal quantum & classical capacity: for all $\varepsilon \in (0,1)$,

$$\lim_{n \rightarrow \infty} \frac{1}{n} Q_{\text{retro}}^{\varepsilon}(\mathcal{N}^{\otimes n}) = \frac{1}{2} \lim_{n \rightarrow \infty} \frac{1}{n} C_{\text{retro}}^{\varepsilon}(\mathcal{N}^{\otimes n}) = \frac{1}{2} (I_{\text{max}}(\mathcal{N}) + I_{\text{doe}}^{\infty}(\mathcal{N}))$$

regularization (non-additive)

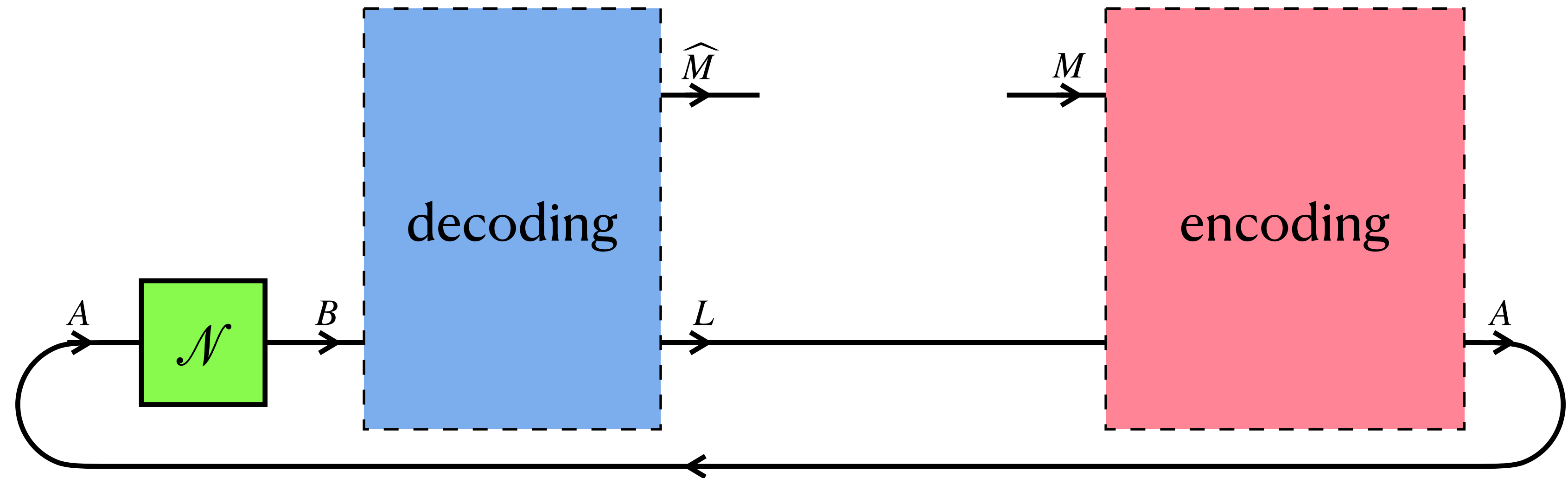
Max-information: $I_{\text{max}}(\mathcal{N}) := \inf_{\lambda, \sigma_B} \left\{ \log_2 \lambda : \mathcal{N}_{A \rightarrow B}[\Phi_{A'A}] \leq \lambda \frac{I_{A'}}{d_{A'}} \otimes \sigma_B \right\}$ König et al., IEEE TIT 55, 4337 (2009)

Doeblin information: $I_{\text{doe}}(\mathcal{N}) := \inf_{\lambda, \tau_B} \left\{ \log_2 \lambda : \lambda \frac{I_{A'}}{d_{A'}} \otimes \tau_B \leq \mathcal{N}_{A \rightarrow B}[\Phi_{A'A}] \right\}$ George et al., arXiv:2503.22823 (2025)

unit-trace Hermitian operator!

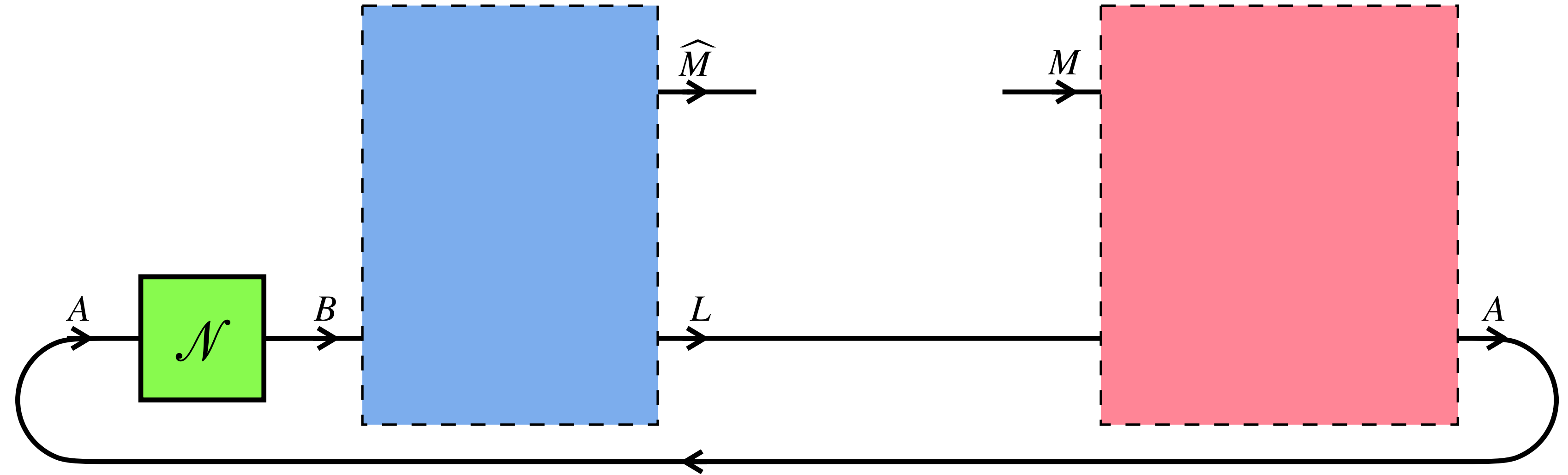
Optimal strategy: Amplified probabilistic teleportation

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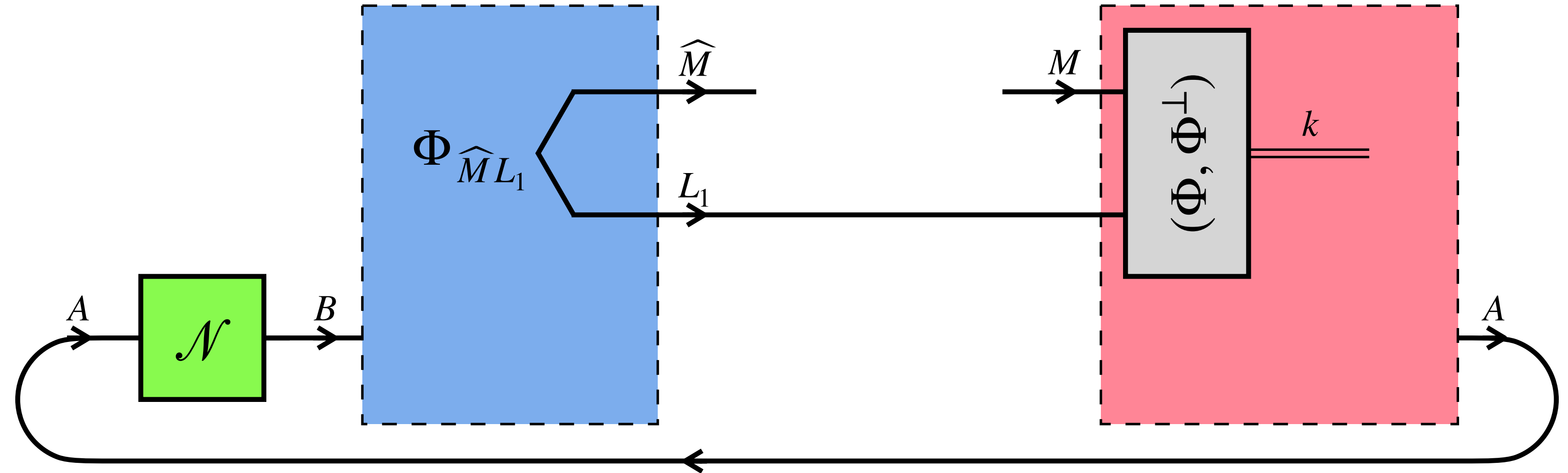
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Instead of encoding the message in the input of the noisy P-CTC, the sender tries to send the message via probabilistic teleportation through $M \rightarrow L_1 \rightarrow \hat{M}$



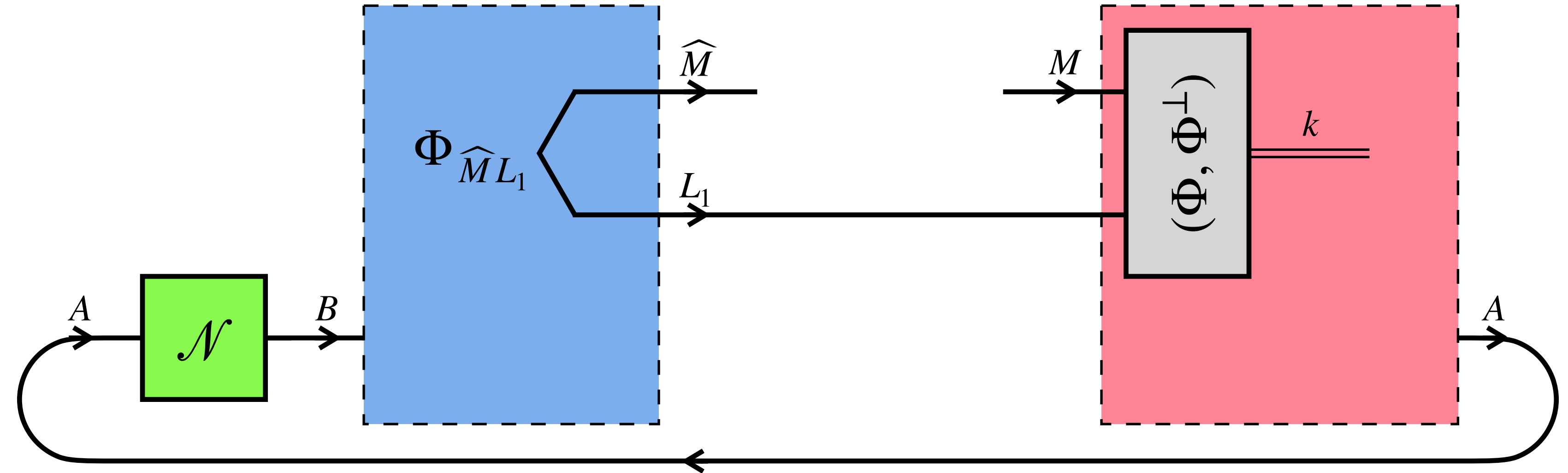
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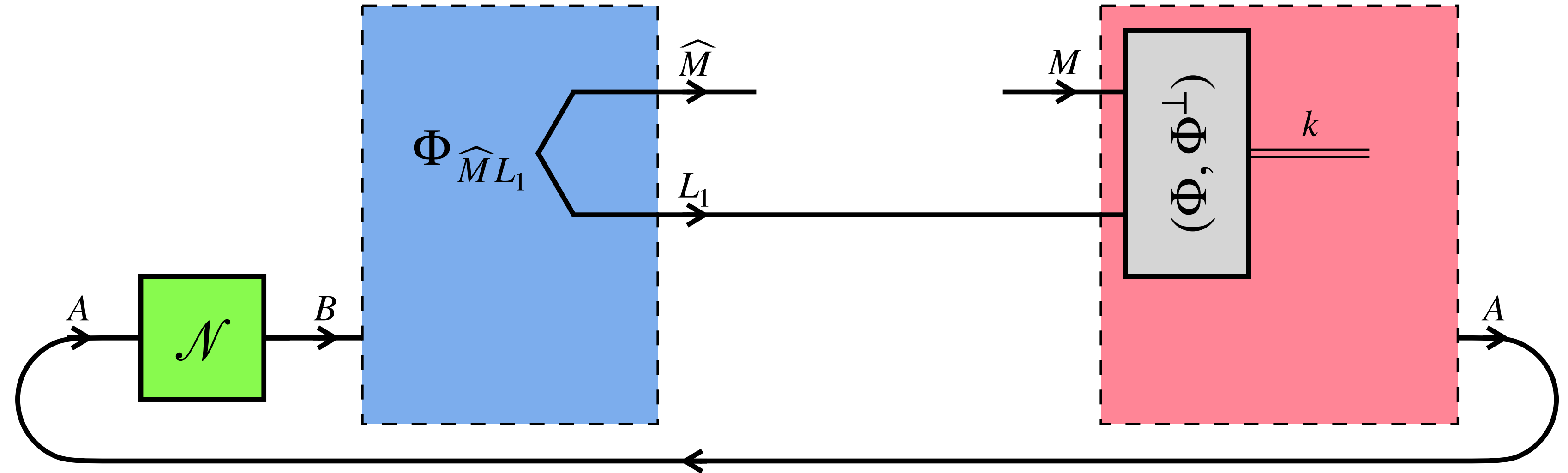
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- The probability of success ($k = 0$) is of course only $1/d_M^2$

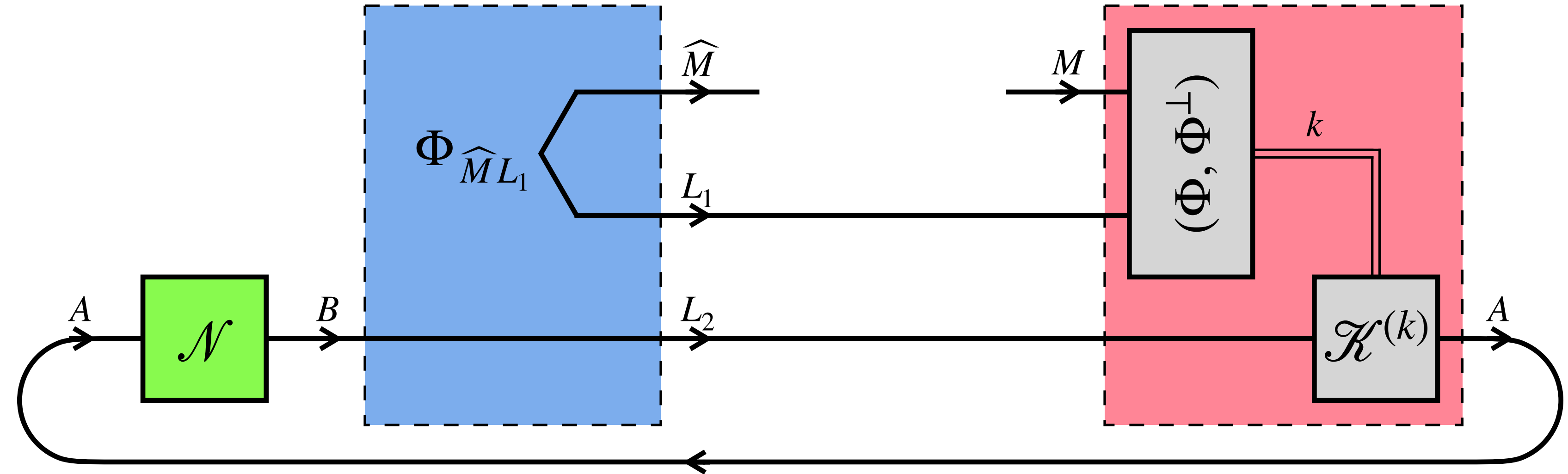
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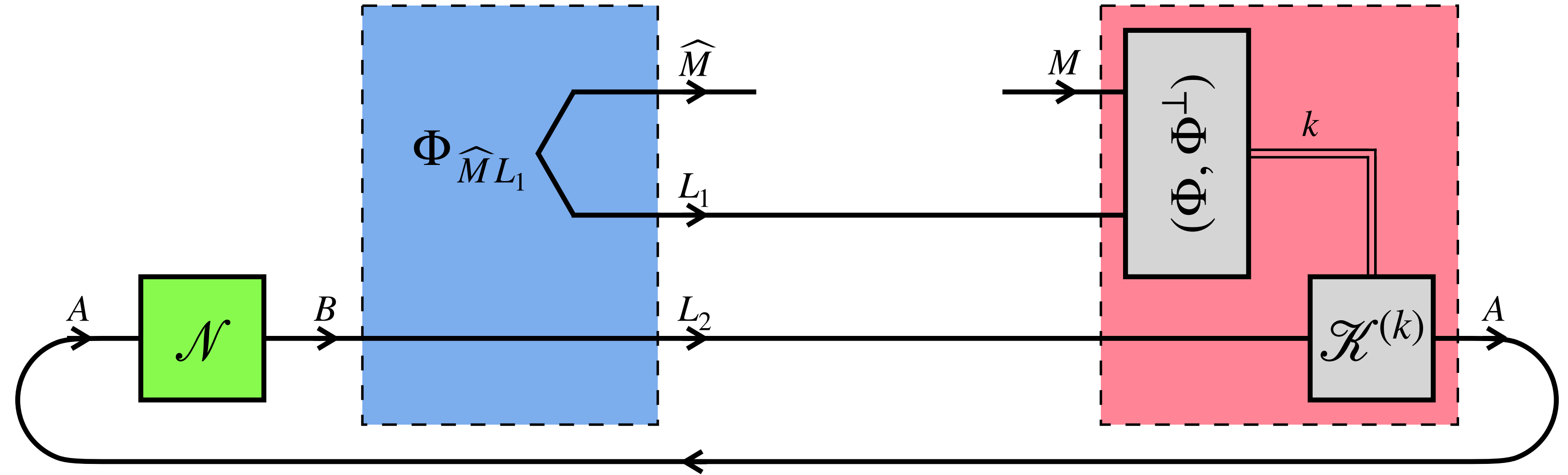
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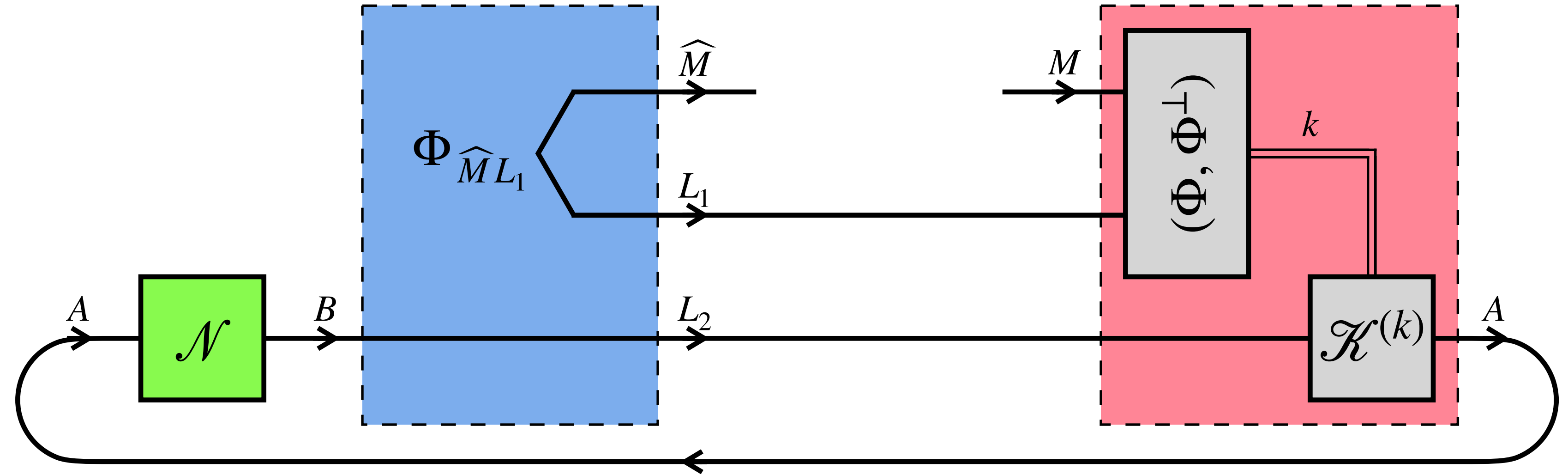
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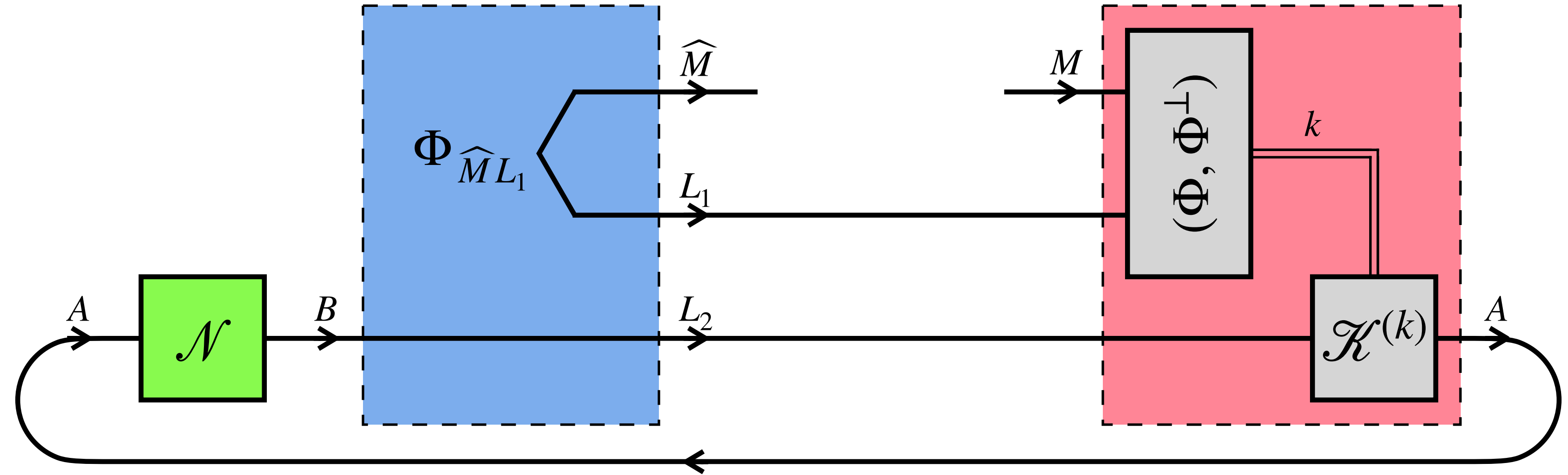


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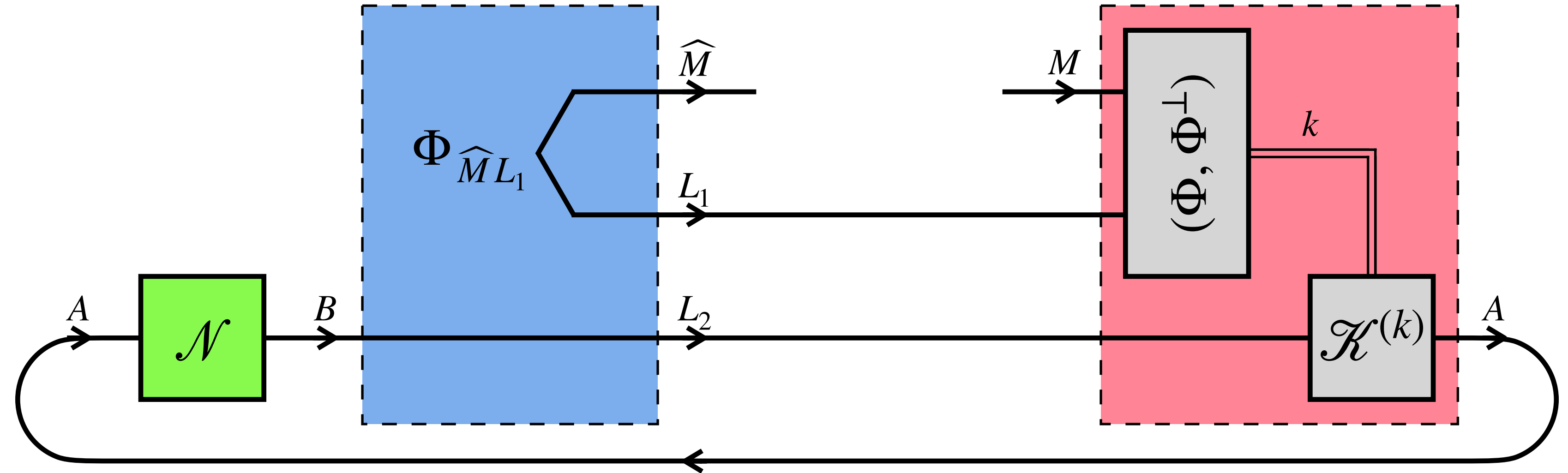


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- When $\log_2 d_M =$ the one-shot retrocausal quantum capacity, the maximized odds $\geq \frac{1 - \varepsilon}{\varepsilon}$

Conclusion

Retrocausal communication: messaging the past from the future through a given noisy P-CTC

- Solved one-shot retrocausal quantum & classical capacity (closed-form, SDP-computable)
- Noisy-“time-machine” coding theorem (backward version of noisy-channel coding theorem)

$$\langle \overleftarrow{\mathcal{N}} \rangle + \infty[q \rightarrow q] \geq \frac{1}{2} (I_{\max}(\mathcal{N}) + I_{\text{doe}}^{\infty}(\mathcal{N})) [q \leftarrow q],$$

noisy time machine noiseless channel noiseless time machine

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Open questions:

- What if the memory that the daughter leaves for the father is noisy?

$$\langle \overleftarrow{\mathcal{N}} \rangle + (?) \langle \overrightarrow{\mathcal{L}} \rangle \geq (?) [q \leftarrow q],$$

- Connection with black-hole final-state models?

Horowitz and Maldacena, JHEP 2004, 008 (2004)

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